

A Perturbative Approach to the Ground State of Interacting Fermions in the Mean Field Limit

Filippo Cozzi

???

Contents

1	Second Quantization	1
1.1	Principles of quantum mechanics	1
1.2	Canonical and grand canonical pictures	6
1.3	One-particle density matrices	15
1.4	Bogolubov transformations and quasi-free pure states	19
1.5	Quadratic hamiltonians	21
2	Perturbation Theory	22
2.1	A toy model: bosonic expectations	22
2.2	Grassmann integrals and fermionic expectations	25
2.3	Perturbative series and Feynman diagrams	29
3	Explicit Calculation at Second Order	30
3.1	Duhamel's formula	30
3.2	Evaluating expectation values	32
3.3	Mean field scaling	38
4	On higher orders of perturbation theory	41
A	More about Hilbert spaces	41
A.1	Special operators	41
A.2	Integration of operator-valued functions	43
A.3	Direct sums and tensor products of Hilbert spaces	45

1 Second Quantization

INTROOOOO

1.1 Principles of quantum mechanics

Preliminaries

In this preliminary section we give an overview of the mathematical objects that appear in the description of quantum systems. The probabilistic nature of a quantum system is realized in the linear structure of a complex vector space that is equipped with an inner product, which produces the probability amplitude for a system prepared in some state to be measured in another. In this setting, linear combinations of state vectors are just probabilistic superpositions. We will give the

necessary definitions regarding a particular class of vector spaces, even though many concepts come from more general settings.

Definition 1.1 (Hilbert space). A (complex) *Hilbert space* $\mathcal{H}$ is a vector space equipped with a sesquilinear map $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ such that $\|\phi\| := \langle \phi, \phi \rangle^{1/2}$ defines a norm which makes $\mathcal{H}$ a complete metric space.

A note on the choice of notation. Sometimes, using Dirac's "bra-ket" notation can help avoid ambiguities. In this notation, a vector would be denoted by $|\psi\rangle \in \mathcal{H}$, a linear functional as $\langle\psi| \in \mathcal{H}^*$, and the inner product (with respect to a symmetric operator, which we haven't yet introduced) as $\langle\psi, A\phi\rangle = \langle\psi|A|\phi\rangle$. An example of such a scenario where this notation *might* be preferred is that of the occupation number basis, which we make use of in Chapter (3).

Definition 1.2. A Hilbert space is said to be *separable* if it admits a countable dense subset.

The relevance of this definition stems from the fact that, if we can find a countable set $\mathcal{S} = \{u_n : n \in \mathbb{N}\}$ of vectors in the Hilbert space $\mathcal{H}$ such that the space of *finite* linear combinations of elements from $\mathcal{S}$ is dense, we have that any vector in $\mathcal{H}$ can be expressed as a limit of such finite linear combinations. That is, the set $\mathcal{S}$ is *complete* (acts as a *basis*) in the sense that, ignoring problems with linear independence, any vector can be identified with a sequence of components $\{a_n : n \in \mathbb{N}\}$, with the condition that the series $\sum_{n \in \mathbb{N}} a_n u_n$ converges (in the norm topology of $\mathcal{H}$). When speaking of Hilbert spaces we will always assume that they are separable.

Lemma 1.1. *If X is a dense subspace of the Hilbert space $\mathcal{H}$, it is always possible to choose a complete orthonormal system $\{u_n : n \in \mathbb{N}\}$ for $\mathcal{H}$ composed only of elements in X .*

Proof. Since $\overline{X} = \mathcal{H}$, for all $x \in \mathcal{H}$ there exists a sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ such that $x_n \rightarrow x$. Let $\{y_n\}_{n \in \mathbb{N}}$ be a complete orthonormal set for $\mathcal{H}$, then for any n there exists a sequence $\{u_m^{(n)}\}_{m \in \mathbb{N}}$ such that $\lim_m u_m^{(n)} = y_n$. Define

$$\mathcal{S} := \{u_m^{(n)} : n, m \in \mathbb{N}\},$$

which is countable because it is a countable union of countable sets. The set $\mathcal{S}$ is complete, indeed let $x \in \mathcal{H}$, then there exists a sequence $\{\lambda_m\}_{m \in \mathbb{N}}$ such that $\sum_{k=1}^{\infty} |\lambda_k|^2 < \infty$ and $x = \sum_{k=1}^{\infty} \lambda_k y_k$: we can express y_k as a limit, as discussed above, to get that

$$x = \sum_{k=1}^{\infty} \lambda_k y_k = \sum_{k=1}^{\infty} \lambda_k \lim_{m \rightarrow \infty} u_m^{(k)} = \lim_{m \rightarrow \infty} \sum_{k=1}^{\infty} \lambda_k u_m^{(k)},$$

which implies that the sequence $\{\sum_{k=1}^N \lambda_k u_m^{(k)}\}_{m, N \in \mathbb{N}} \subseteq X$ converges to x (you sure?). Lastly, recall the infinite-dimensional version of the Gram-Schmidt orthonormalization procedure, then it is clear that to $\mathcal{S}$ we can associate a complete orthonormal system. $\square$

In quantum mechanics, observables are represented by a special type of linear map on the state space, so let us briefly recall the (purely formal) motivation behind this. If A is a physical observable, the result of an experiment is the collapse of the system onto a state associated to a well-defined value of A . A generic state ψ will be a superposition of such vectors, which we may call $\{\psi_i : i \in \mathbb{N}\}$, and let us call $\{\lambda_i : i \in \mathbb{N}\} \subset \mathbb{R}$ the well-defined values of A associated to such vectors. The generic state ψ then encodes the probability amplitudes for system to collapse in each ψ_i , which suggests that the statistical expectation for the measurement of A , when the system is in the state $\psi = \sum_i \lambda_i \psi_i$, would read something like

$$\langle A \rangle_{\psi} = \sum_{i \in I} \lambda_i \langle \psi_i, \psi_i \rangle = \langle \psi, A \psi \rangle$$

if we could represent the observable by a linear map $A : \mathcal{H} \rightarrow \mathcal{H}$. Let us make this precise and give meaning to what we just wrote.

Definition 1.3 (Operator). A *linear operator* (or simply an *operator*) A on $\mathcal{H}$ is a linear map $A : D(A) \rightarrow \mathcal{H}$ defined on a subspace $D(A) \subseteq \mathcal{H}$. If $D(A)$ is dense in $\mathcal{H}$, A is said to be *densely defined*.

We will often write “operator” to mean a densely defined operator on a Hilbert space, unless stated otherwise. Moreover, note that the domain on which an operator acts is part of the definition, and will therefore need to be specified case by case.

Definition 1.4 (Extension of an operator). If A and B are two operators such that $D(A) \subseteq D(B)$ and $A\phi = B\phi$ for all $\phi \in D(A)$, then we say that B *extends* (or is an extension of) A , and we write $A \subseteq B$.

Definition 1.5 (Bounded operator). A densely defined operator A is said to be *bounded* if it is continuous with respect to the norm topology, and *unbounded* if it is not bounded.

Remark 1. By linearity, continuity is equivalent to the condition that

$$\|A\| := \sup_{\substack{\psi \in D(A) \\ \|\psi\|=1}} \|A\psi\| < \infty,$$

and the quantity $\|A\|$ is called the *operator norm* of A .

Proposition 1.2. *Given $A : D(A) \rightarrow \mathcal{H}$ a densely defined bounded operator on $\mathcal{H}$, there exists a unique extension to a bounded operator on the entire space $\tilde{A} : \mathcal{H} \rightarrow \mathcal{H}$.*

Proof. Let $\{x_n\}_{n \in \mathbb{N}} \subseteq D(A)$ be a Cauchy sequence in $D(A)$, then there exists a unique $x \in \mathcal{H}$ such that $x_n \rightarrow x$. Because A is bounded it holds that $\|Ax_n - Ax_m\| \leq \|A\| \|x_n - x_m\|$ and therefore the sequence $\{Ax_n\}_{n \in \mathbb{N}}$ is also a Cauchy sequence. As such, there exists a unique $y \in \mathcal{H}$ such that $Ax_n \rightarrow y$. Define the operator $\tilde{A}$ as

$$\begin{aligned} \tilde{A} : \mathcal{H} &\longrightarrow \mathcal{H} \\ x &\longmapsto \tilde{A}x := \lim_n Ax_n \end{aligned}$$

where $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence that converges to x . $\tilde{A}$ extends A and is defined on the entire space. Moreover, it is bounded as $\|A\| < \infty$ and

$$\|\tilde{A}x\| = \|\lim_n Ax_n\| = \lim_n \|Ax_n\| \leq \lim_n \|A\| \|x_n\| = \|A\| \|x\|.$$

□

It is worth underlying that from Proposition (1.2) follows that for bounded operators it is possible to deal with dense subspaces without losing generality. In particular, since we are always assuming the Hilbert space to be separable, it will suffice to define the image of only the basis elements.

Definition 1.6 (Adjoint). If A is an operator on $\mathcal{H}$ we define the *adjoint* A^* of A as the operator $A^* : D(A^*) \rightarrow \mathcal{H}$ defined on

$$D(A^*) = \{\phi \in \mathcal{H} : \sup_{\substack{\psi \in D(A) \\ \|\psi\|=1}} |\langle \phi, A\psi \rangle| < \infty\}$$

and with $A^*\phi$ defined for all $\phi \in D(A^*)$ as

$$\langle A^*\phi, \psi \rangle = \langle \phi, A\psi \rangle \quad \forall \psi \in D(A).$$

If $A^* = A$ then A is said to be *self-adjoint*.

Definition 1.7 (Observable). A *physical observable* (or simply *observable*) is a self-adjoint operator A on the Hilbert space $\mathcal{H}$ of physical states. Given a normalized vector $\psi \in \mathcal{H}$ we define the *expectation of A on ψ* as

$$\langle A \rangle_\psi := \langle \psi, A\psi \rangle.$$

Remark 2. From the self-adjointness of A we have that $\langle \psi, A\phi \rangle = \langle A\psi, \phi \rangle$ for all $\phi, \psi \in \mathcal{H}$, which is equivalent to saying that $\langle \psi, A\psi \rangle \in \mathbb{R}$ for all $\psi \in \mathcal{H}$, i.e. expectation values of A must be real numbers.

Proposition 1.3. *The adjoint of a bounded operator is a bounded operator.*

Proof. **CHECK.** We are only considering densely defined operators, therefore, without loss of generality, let $A : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded operator on the whole space $\mathcal{H}$. For all $\phi \in D(A^*)$, the map $\psi \mapsto \langle A^*\phi, \psi \rangle$ is a linear functional on $\mathcal{H}$, where

$$D(A^*) = \left\{ \phi \in \mathcal{H} : \sup_{\|\psi\|=1} \langle \phi, A\psi \rangle < \infty \right\}.$$

Then, by Schwarz's inequality, we have that for every $\phi \in D(A^*)$

$$\|A^*\phi\| = \sup_{\|\psi\|=1} |\langle A^*\phi, \psi \rangle| = \sup_{\|\psi\|=1} |\langle \phi, A\psi \rangle| \leq \sup_{\|\psi\|=1} \|A\psi\| \|\phi\|,$$

where the first step makes use of Riesz representation theorem, which states that the norm of the vector $A^*\phi$ is equal to the norm of the functional we previously wrote. It follows from the boundedness of A that

$$\|A^*\phi\| \leq \|\phi\| \|A\| < \infty$$

which implies that A^* is bounded. □

Remark 3. We can introduce a partial ordering on self-adjoint operators. If A and B are self-adjoint and defined on a common domain $D \subseteq \mathcal{H}$, then we say that $A < B$ if $\langle \phi, A\phi \rangle < \langle \phi, B\phi \rangle$ for all $\phi \in D$.

We point to Appendix A.1 for a discussion on the most important definitions and properties of special types of operators, and end here the introduction of the basic objects used. A key role in the discussion of quantum systems is played by the space of square integrable functions on $\mathbb{R}^n$

$$L^2(\mathbb{R}^n) = \left\{ f : \mathbb{R}^n \rightarrow \mathbb{C} \mid \left(\int_{\mathbb{R}^n} dx |f(x)|^2 \right)^{1/2} < \infty \right\}$$

with $\langle f, g \rangle := \int_{\mathbb{R}^n} dx \overline{f(x)} g(x),$

whose elements are often called wavefunctions. Its relevance stands from the fact, that in this space, the squared norm of a state $\psi \in L^2(\mathbb{R}^n)$ makes it possible to write probability amplitudes as integrals of a *probability density*, given as the point-wise modulus squared of the wave function: $|\psi(x)|^2$.

Going back to the definition of expectation value of an observable, we can now make an important generalization. Suppose the system has probability p_i of being prepared in some initial state ψ_i , of course given the normalization condition $\sum_i p_i = 1$, then the expectation of an observable A would be the weighted average of the expectations on each initial state.

Definition 1.8 (Pure and mixed states). A *pure state* is a vector in $\mathcal{H}$, while a *mixed state* is a statistical mixture of pure states given by the collection of probability-state pairs $\{(p_i, \psi_i) : i \in \mathbb{N}\}$, with $\sum_i p_i = 1$. If A is an observable, the expectation of A on a mixed state is defined as

$$\langle A \rangle = \sum_{i \in \mathbb{N}} p_i \langle \psi_i, A\psi_i \rangle.$$

We can take this definition further by following another formal argument. Assuming the observable A is a compact self-adjoint operator, by the spectral theorem we can write it as a series of projectors onto its eigenspaces. Let $\{\phi_\lambda : \lambda \in \sigma(A)\}$ be the eigenvectors of A , then

$$A = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \cdot \rangle \phi_\lambda.$$

Using this, we write the expectation of A on the mixed state $\{(p_i, \psi_i)\}$ as

$$\langle A \rangle = \sum_{i \in I} p_i \langle \psi_i, A \psi_i \rangle = \sum_{i \in \mathbb{N}, \lambda \in \sigma(A)} p_i \langle \psi_i, \lambda \langle \phi_\lambda, \psi_i \rangle \phi_\lambda \rangle = \sum_{\lambda \in \sigma(A)} \lambda \langle \phi_\lambda, \rho \phi_\lambda \rangle = \text{Tr}(\rho A)$$

where in the last step we defined the *density matrix* which uniquely identifies the mixed state

$$\rho := \sum_i p_i \langle \psi_i, \cdot \rangle \psi_i.$$

This argument brings us to a general definition of a quantum state. Recall that $\mathfrak{B}(\mathcal{H})$ is the C^* -algebra of bounded operators on $\mathcal{H}$ (see Appendix A.1).

Definition 1.9. A general quantum state is an operator ρ on $\mathcal{H}$ such that

- $\text{Tr}(\rho) = 1$ (probability normalization condition)
- $\rho^* = \rho$ (self-adjointness)
- $\rho \geq 0$ (positivity of probabilities)

and it defines the *expectation functional*

$$\begin{aligned} \langle \cdot \rangle_\rho : \mathfrak{B}(\mathcal{H}) &\longrightarrow \mathbb{C} \\ A &\longmapsto \langle A \rangle_\rho := \text{Tr}(\rho A). \end{aligned}$$

There is a correspondence between the set of mixed states (pure states are just a special case) and the set of quantum states (i.e. a subset of the dual of $\mathfrak{B}(\mathcal{H})$). We are interested in the state the system is found in when at thermal equilibrium, and from statistical mechanics we know this state is such that the energy spectrum is populated according to Boltzmann's distribution $p(\epsilon) = e^{-\beta\epsilon}/Z$. With the aforementioned correspondence we may identify such a state *at a given temperature* with the mixture $\{(e^{-\beta\epsilon_j}/Z, \psi_j)\}$ if ψ_j are energy eigenstates. To make this precise we need to properly define the fundamental thermodynamic quantities required, which also brings us to the main focus of our work: the free energy, which we will define starting from the known thermodynamic relation $F = U - TS$.

Definition 1.10 (Von Neumann entropy). In the same notation used so far, the *Von Neumann entropy* of a state $\langle \cdot \rangle_\rho$ is defined as

$$S(\langle \cdot \rangle_\rho) := - \sum_{n=1}^{\infty} p_n \log(p_n),$$

where we are using the convention that $t \log(t) = 0$ if $t = 0$.

Definition 1.11 (Stability and ground state). A physical system governed by a Hamiltonian, i.e. energy operator, H on $\mathcal{H}$ is said to be *stable* if

$$\inf_{\phi \in D(H), \|\phi\|=1} \langle \phi, H \phi \rangle > -\infty.$$

We define the *ground state energy* as

$$E = \inf_{\phi \in D(H), \|\phi\|=1} \langle \phi, H\phi \rangle,$$

and the *ground state, if it exists*, as a (normalized) minimizer for the energy expectation, i.e. a unit vector $\psi_0 \in D(H)$ such that $\langle \psi_0, H\psi_0 \rangle = E$.

Remark 4. If the ground state ψ_0 exists and the ground state energy is E then it holds that $H\psi_0 = E\psi_0$, i.e. ψ_0 is an eigenvector of the energy operator with eigenvalue equal to the ground state energy.

Definition 1.12 (Free energy). The *free energy* of a stable system at temperature $T \geq 0$ is defined as

$$F(T) := \inf_{\substack{\langle \cdot \rangle_\rho \\ \langle H \rangle_\rho < \infty}} (\langle H \rangle - T S(\langle \cdot \rangle)),$$

which is possibly $-\infty$. If, for $T > 0$, there exists a minimizer for the free energy, then the minimizing state is called a *Gibbs state at (inverse) temperature β* .

Gibbs states are thermal equilibrium states, and for $\beta < \infty$ they are generally mixed, while at $\beta = \infty$ we find the, usually pure, ground state. We will denote Gibbs states by $\langle \cdot \rangle_\beta$, despite the slight ambiguity between this notation and the one used for a generic mixed state, and we will also specify the Hamiltonian, when needed, by writing $\langle \cdot \rangle_{\beta, H}$.

Remark 5. A system composed of a single free particle with state space $L^2(\mathbb{R})$ and governed by the Hamiltonian $H = -\nabla^2$ is stable but it does not admit a ground state, and $F(T) = -\infty$ for all $T > 0$. However, this Hamiltonian does admit a ground state when restricting the state space to wavefunctions on a compact interval with periodic boundary conditions. The stability of free particles is essential, as will become clear in the third chapter.

Proposition 1.4. If $\langle \cdot \rangle_\beta$ is a Gibbs state at $\beta < \infty$ (i.e. at $T > 0$), then for any bounded self-adjoint operator A we have that

$$\langle A \rangle = \frac{\text{Tr}(Ae^{-\beta H})}{\text{Tr}(e^{-\beta H})}.$$

In particular, $e^{-\beta H}$ is a trace class operator.

Proof. **IMPORTANT** (and maybe add discussion of $e^{\pm \tau H_0}$ and of $Ve^{-\tau H_0}$ where V is the potential). □

We refer to the appendix for the definition of trace class operators and point out that such operators are compact, which makes $e^{-\beta H}$ a very well-behaved operator in computations.

1.2 Canonical and grand canonical pictures

So far we talked about a generic (separable) Hilbert space, but for the most part we will deal with Hilbert spaces constructed in a specific way: tensor products, which are the natural spaces in which to set physical systems composed of multiple degrees of freedom. We are particularly interested in two special subspaces of a tensor product that encode the *statistics* of the particles considered, a property that describes indistinguishability between particles in a quantum system. Once again we point to Appendix (A.3) for a discussion on the construction of tensor products (and direct sums) of Hilbert spaces.

Consider a system of N particles whose physical states belong to the spaces $\mathfrak{h}_1, \dots, \mathfrak{h}_N$. The possible states of this system are described by the tensor product

$$\mathcal{H}_N = \mathfrak{h}_1 \otimes \dots \otimes \mathfrak{h}_N.$$

Given the energy operators for each particle $h_1, \dots, h_N$, we lift these operators to the space $\mathcal{H}_N$ with the identification

$$h_i \longmapsto \mathbb{1} \otimes \dots \otimes h_i \otimes \dots \otimes \mathbb{1}.$$

The total energy of the system, given only the “internal” degrees of freedom for each particle and disregarding any interaction, is naturally given by

$$H_N^{\text{in}} = h_1 + \dots + h_N \quad (1)$$

with

$$D(H_N^{\text{in}}) = \text{span}\{\phi_1 \otimes \dots \otimes \phi_N : \phi_i \in D(h_i), i = 1, \dots, N\}$$

which is dense in $\mathcal{H}_N$ if $D(h_i)$ is dense in $\mathfrak{h}_i$ for all i .

Theorem 1.5 (ground state of non-interacting (distinguishable) particles). *If the ground state energies of the Hamiltonians $h_1, \dots, h_N$ are*

$$e_j = \inf_{\phi \in D(h_j), \|\phi\|=1} \langle \phi, h_j \phi \rangle, \quad j = 1, \dots, N$$

then the ground state energy of H_N^{in} is $\sum_{j=1}^N e_j$. Moreover, if $\phi_1, \dots, \phi_N$ are the ground states of these same Hamiltonians, then the ground state of H_N^{in} is $\phi_1 \otimes \dots \otimes \phi_N$.

Proof. If $\Psi \in D(H_N^{\text{in}})$ is a unit vector, then we may write $\Psi = \psi_1 \otimes \Psi_1 + \dots + \psi_K \otimes \Psi_K$, where $\psi_1, \dots, \psi_K \in D(h_1)$ and $\Psi_1, \dots, \Psi_K \in \mathfrak{h}_2 \otimes \dots \otimes \mathfrak{h}_N$ are orthonormal. Since Ψ is normalized this implies that $\|\psi_1\|^2 + \dots + \|\psi_K\|^2 = 1$, thus

$$\langle \Psi, h_1 \Psi \rangle = \sum_{i=1}^K \langle \psi_i, h_1 \psi_i \rangle \geq \sum_{i=1}^K \|\psi_i\|^2 e_1 = e_1.$$

This implies that $\langle \Psi, H_N^{\text{in}} \Psi \rangle \geq \sum_{j=1}^N e_j$. Conversely, for every $\varepsilon > 0$ we can choose unit vectors $\phi_j \in D(h_j)$ such that $\langle \phi_j, h_j \phi_j \rangle \leq e_j + \varepsilon$ with $j = 1, \dots, N$, which implies that, for $\Psi = \phi_1 \otimes \dots \otimes \phi_N$ (still a unit vector),

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{j=1}^N \langle \phi_j, h_j \phi_j \rangle \leq \sum_{j=1}^N e_j + N\varepsilon,$$

which holds if, and only if, $\langle \Psi, H_N^{\text{in}} \Psi \rangle \leq \sum_{j=1}^N e_j$, thus we proved the equality. Furthermore, it is clear that if $\phi_1, \dots, \phi_N$ are the respective ground state eigenvectors of the N one-body Hamiltonians, their tensor product is an eigenvector of H_N^{in} with eigenvalue given precisely by $\sum_{j=1}^N e_j$, the ground state energy. $\square$

If we consider interactions we need to lift additional operators. For pair-wise interactions (but this procedure can be easily generalized to N -body interactions) to each pair (i, j) of distinct particles ($i \neq j$) we associate an operator W_{ij} on a suitable domain. To lift this operator, consider the vectors $\phi_i \in \mathfrak{h}_i$ and identify W_{ij} with the operator on $\mathcal{H}_N$ that acts as

$$W_{ij}(\phi_1 \otimes \dots \otimes \phi_i \otimes \dots \otimes \phi_j \otimes \dots \otimes \phi_N) = \phi_1 \otimes \dots \otimes \psi_i \otimes \dots \otimes \psi_j \otimes \dots \otimes \phi_N$$

in the case where $W_{ij}\phi_i \otimes \phi_j = \psi_i \otimes \psi_j$ (and express the image as a sum of simple tensors in the case where it is not factorizable as one). Without specifying the correct domain we arrive at the formal way of writing the Hamiltonian of a system of N pair-wise interacting particles:

$$H_N = H_N^{\text{in}} + \sum_{1 \leq i < j \leq N} W_{ij} = \sum_{j=1}^N h_j + \sum_{1 \leq i < j \leq N} W_{ij}. \quad (2)$$

Requiring that particles be indistinguishable from one another means that both their internal energy operators and the pair-wise interactions do not depend on the specific particle considered. This means that

$$\begin{aligned} \mathfrak{h}_1 &= \dots = \mathfrak{h}_N =: \mathfrak{h}, \\ h_1 &= \dots = h_N =: h, \\ W_{ij} &= W_{kl} =: W \quad \forall i \neq j, k \neq l. \end{aligned}$$

Indistinguishability also requires an additional property that physical states must satisfy, so let us take the time to recall a formal explanation of what this property is. Consider the simplest case of a system of two indistinguishable particles in the states $\psi, \phi \in \mathfrak{h}$. Because quantum states can be told apart only through probability amplitudes, two vectors that differ only by an overall phase can be considered indistinguishable, as $|\langle \phi, e^{i\theta} \psi \rangle|^2 = \|\phi\|^2$ no matter the phase θ . Therefore we can consider an exchange operator that flips the order in a simple tensor in $\mathfrak{h} \otimes \mathfrak{h}$

$$\text{Ex}(\phi \otimes \psi) = \psi \otimes \phi$$

and say that two indistinguishable particles only admit physical states that differ at most by an overall phase from their exchanged version, i.e. there must exist some α such that

$$\text{Ex}(\phi \otimes \psi) = e^{i\alpha} \phi \otimes \psi.$$

But since Ex^2 should act as the identity, as flipping two times the order of the particle states results in the original state, we can conclude that

$$\text{Ex}^2 = \mathbb{1} \implies e^{2i\alpha} = 1 \implies \alpha = \pm 1.$$

The sign $\alpha = \pm 1$ is what determines the statistics of the two identical particles: (+) for *bosons* and (-) for *fermions*. Generalizing to the case of N identical particles (of the same statistics) means defining physical states as those that belong to the $+1$ or -1 joint eigenspaces of *all* the exchange operators between two of the N particles, i.e. the subspace of completely symmetric or completely anti-symmetric tensors respectively.

We will now make precise the notion of bosonic and fermionic physical states. On the space $\mathcal{H}_N$ we can define a natural action for the group of permutations S_N . Indeed, we can build the unitary representation U_σ of $\sigma \in S_N$ on this space by specifying the action on the usual basis (see Appendix A.3):

$$U_\sigma(u_1 \otimes \dots \otimes u_N) := u_{\sigma^{-1}(1)} \otimes \dots \otimes u_{\sigma^{-1}(N)}.$$

Define the operators

$$\begin{aligned} P_+ &:= \frac{1}{N!} \sum_{\sigma \in S_N} U_\sigma \\ P_- &:= \frac{1}{N!} \sum_{\sigma \in S_N} \text{sgn}(\sigma) U_\sigma. \end{aligned} \quad (3)$$

These are unitary by construction and **one can show** that they are also projections.

Definition 1.13 (Symmetric and anti-symmetric tensor products). Given $\mathfrak{h}$ the one-particle Hilbert space, the *symmetric tensor product* is the subspace

$$\bigotimes_{sym}^N \mathfrak{h} := P_+ \left(\bigotimes^N \mathfrak{h} \right) \subset \mathcal{H}_N$$

while the *anti-symmetric tensor product* is the subspace

$$\bigwedge^N \mathfrak{h} := P_- \left(\bigotimes^N \mathfrak{h} \right) \subset \mathcal{H}_N.$$

We also define the anti-symmetric tensor product of vectors as

$$u_1 \wedge \cdots \wedge u_N = \sqrt{N!} P_- (u_1 \otimes \cdots \otimes u_N).$$

Bosonic physical states belong in the symmetric tensor product, while fermionic physical states belong in the anti-symmetric tensor product.

Remark 6. In the special case where $\mathfrak{h} = L^2(\mathbb{R}^n)$, anti-symmetric tensor products between wave-functions can be constructed as Slater determinants, i.e.

$$u_1 \wedge \cdots \wedge u_N(x_1, \dots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} u_1(x_1) & \cdots & u_N(x_1) \\ \vdots & & \vdots \\ u_1(x_N) & \cdots & u_N(x_N) \end{pmatrix}$$

Remark 7. If $\dim(\mathfrak{h}) < N$ then $\bigwedge^N \mathfrak{h} = \{0\}$, i.e. the anti-symmetrization of N vectors in a space of dimension less than N is always the null vector. Thus, if the Hilbert space of a single fermionic particle is finite-dimensional, the allowed number of particles in the canonical and grand canonical pictures will be finite.

With the restrictions we put on the N -body Hamiltonian, it is clear that H_N maps the spaces $\bigotimes_{sym}^N \mathfrak{h}$ and $\bigwedge^N \mathfrak{h}$ into themselves, which means that we can restrict this N -body Hamiltonian to either of the two subspaces and get a well-defined operator (provided that a suitable domain for the starting Hamiltonian is chosen). The operators $P_+(H_N)$ and $P_-(H_N)$ are respectively called the *bosonic* and *fermionic* N -body Hamiltonians for interacting indistinguishable particles.

This way of defining the energy of the system characterises the *canonical picture*, where the number of particles is fixed. But in statistical mechanics, allowing the number of particles to vary can actually make things easier. Allowing N to vary defines the *grand canonical picture*, and we shall now discuss its state space.

Definition 1.14 (Fock space). For every $N \in \mathbb{N}$, let $\mathfrak{h}_1, \dots, \mathfrak{h}_N$ be one-particle Hilbert spaces, the *Fock space* is defined as

$$\mathcal{F} := \bigoplus_{N=0}^{\infty} \bigotimes_{i=1}^N \mathfrak{h}_i$$

where, as a convention, we interpret $\mathfrak{h}_1 \otimes \cdots \otimes \mathfrak{h}_N$ for $N = 0$ to be just $\mathbb{C}$. The state $1 \in \mathbb{C}$ is called the *vacuum state* and often denoted $|0\rangle$, $|\Omega\rangle$ or Ω .

On this space, the energy operator is naturally given by the sum of all possible N -body Hamiltonians:

$$H = \bigoplus_{N=0}^{\infty} H_N, \quad \text{with } H \bigoplus_{N=0}^{\infty} \Psi_N = \bigoplus_{N=0}^{\infty} H_N \Psi_N,$$

with H_0 taken to be the null operator (which means we are viewing the vacuum as having zero energy), and its domain is

$$D(H) = \left\{ \Psi = \bigoplus_{N=0}^{\infty} \Psi_N \in \mathcal{F} : \Psi_N \in D(H_N), \sum_{N=0}^{\infty} \|H_N \Psi_N\|^2 < \infty \right\}.$$

Introducing indistinguishability in this picture means repeating the steps we took for the N-body case in each N -particle sector, which gets us to the definition of the *fermionic* and *bosonic* Fock spaces. Given $\mathfrak{h}$ the state space for a single particle, these spaces are, respectively

$$\begin{aligned} \mathcal{F}^F(\mathfrak{h}) &:= \bigoplus_{N=0}^{\infty} \bigwedge^N \mathfrak{h} \\ \mathcal{F}^B(\mathfrak{h}) &:= \bigoplus_{N=0}^{\infty} \bigotimes_{\text{sym}}^N \mathfrak{h}. \end{aligned}$$

One of the most powerful aspects of the grand canonical picture is that all observables of the system can be written in the same way using special operators called *creation/annihilation operators*. Starting from the Fock space $\mathcal{F} = \bigoplus_{N=0}^{\infty} \bigotimes^N \mathfrak{h}$ we define operators that add or remove a single particle state $f \in \mathfrak{h}$ as

$$\begin{aligned} a(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N} \langle f, f_1 \rangle_{\mathfrak{h}} f_2 \otimes \cdots \otimes f_N, \\ a^*(f)(f_1 \otimes \cdots \otimes f_N) &:= \sqrt{N+1} f \otimes f_1 \otimes \cdots \otimes f_N. \end{aligned} \quad (4)$$

Here, for every $f \in \mathfrak{h}$, $a(f)$ is called *annihilation operator* and $a^*(f)$ is called *creation operator*. Note the very important special case where an annihilation operator acts on the vacuum: $a(f)|0\rangle = 0$ for all $f \in \mathfrak{h}$, i.e. annihilation operators annihilate the vacuum. To indicate an operator that could be either of the two, we write $a^{\#}(f)$. With this definition the domain would be the set of pure tensors, but by linearity we can extend this to the space $\cup_{M=0}^{\infty} \bigoplus_{N=0}^M \mathcal{H}_N$, which is dense in $\mathcal{F}$.

Proposition 1.6. *Creation and annihilation are the adjoint of one another, i.e. for all $f \in \mathfrak{h}$ and for all $\Psi, \Phi \in \cup_{M=0}^{\infty} \bigoplus_{N=0}^M \mathcal{H}_N$ it holds that*

$$\langle a(f)\Psi, \Phi \rangle_{\mathcal{F}} = \langle \Psi, a^*(f)\Phi \rangle_{\mathcal{F}}.$$

Proof. It suffices to prove the equality for simple tensors, so let $\psi_1 \otimes \cdots \otimes \psi_N \in \bigotimes^N \mathfrak{h}$ and $\phi_0 \otimes \phi_1 \otimes \cdots \otimes \phi_N \in \bigotimes^N \mathfrak{h}$. Then, for all $f \in \mathfrak{h}$,

$$\begin{aligned} \langle \psi_1 \otimes \cdots \otimes \psi_N, a(f)\phi_0 \otimes \cdots \otimes \phi_N \rangle &= \sqrt{N+1} \langle f, \phi_0 \rangle \langle \psi_1 \otimes \cdots \otimes \psi_N, \phi_1 \otimes \cdots \otimes \phi_N \rangle \\ &= \sqrt{N+1} \langle f, \phi_0 \rangle \prod_{j=1}^N \langle \psi_j, \phi_j \rangle, \end{aligned}$$

and similarly

$$\langle a^*(f)\psi_1 \otimes \cdots \otimes \psi_N, \phi_0 \otimes \cdots \otimes \phi_N \rangle = \sqrt{N+1} \langle f, \phi_0 \rangle \prod_{j=1}^N \langle \psi_j, \phi_j \rangle.$$

□

Once again we introduce indistinguishability into the picture. Looking at the definition of these operators, it is clear that if a state $\Psi_N \in D(a^{\#}(f))$ is completely symmetric or completely anti-symmetric then applying an annihilation operator produces a state that is still completely symmetric

or anti-symmetric (as Ψ_N can be written as a linear combination of all $N!$ permutations of N states and thus $a(f)$ removes every particle exactly once), but applying a creation operator does introduces a state that is not yet accounted for in the (anti-)symmetrization. We then define *bosonic* (+) and *fermionic* (-) creation/annihilation operators as the projection onto the correct subspaces:

$$a_{\pm}(f) := a(f), \quad a_{\pm}^*(f) := P_{\pm}(a^*(f)) \quad \forall f \in \mathfrak{h}.$$

Remark 8. Bosonic and fermionic creation/annihilation operators are still densely defined and adjoints of one another (within their respective statistics).

Remark 9. The map $\mathfrak{h} \ni f \mapsto a_{\pm}^*(f)$ is linear but $\mathfrak{h} \ni f \mapsto a_{\pm}(f)$ is conjugate linear.

Remark 10. The normalization we used in (4) changes slightly when projecting the image onto the (anti-)symmetric subspaces. Indeed **it can be shown that**

$$\begin{aligned} a_{\pm}(f)P_{\pm}(f_1 \otimes \cdots \otimes f_N) &= \sqrt{N+1} \sum_{\sigma \in S_N} \langle f, f_{\sigma(1)} \rangle f_{\sigma(2)} \otimes \cdots \otimes f_{\sigma(N)}, \\ a_{\pm}^*(f)f_1 \otimes \cdots \otimes f_N &= \frac{1}{\sqrt{N}} \sum_{\sigma \in S_N} \sum_{j=1}^{N+1} f_{\sigma(1)} \otimes \cdots \otimes \underset{(j\text{-th})}{f} \otimes \cdots \otimes f_{\sigma(N)}. \end{aligned}$$

Recall the definition of the commutator and anticommutator between two operators defined on a common domain: $\{A, B\} := AB + BA$ and $[A, B] := AB - BA$.

Proposition 1.7 (CCR/CAR). *The operators a_- and a_-^* satisfy the Canonical Commutation Relations (CCR), while the operators a_+ and a_+^* satisfy the Canonical Anticommutation Relations (CAR), i.e. for all $f, g \in \mathfrak{h}$ it holds that*

$$\begin{aligned} \{a_+(f), a_+^*(g)\} &= \langle f, g \rangle_{\mathfrak{h}}, & \{a_+(f), a_+(g)\} &= \{a_+^*(f), a_+^*(g)\} = 0 \\ [a_-(f), a_-^*(g)] &= \langle f, g \rangle_{\mathfrak{h}}, & [a_-(f), a_-(g)] &= [a_-^*(f), a_-^*(g)] = 0 \end{aligned}$$

Proof. **EXPLICIT CALCULATION?** □

Remark 11. Fermionic creation/annihilation operators are bounded. To see this, let $f \in \mathfrak{h}$, it is clear that for all normalized $\Psi \in D(a(f))$ it holds that

$$\begin{aligned} \|a_-(f)\Psi\|^2 &= \langle \Psi, a_-^*(f)a_-(f)\Psi \rangle = -\langle \Psi, a_-(f)a_-^*(f)\Psi \rangle + \langle \Psi, \{a_-(f), a_-^*(f)\}\Psi \rangle \\ &= -\langle a_-^*(f)\Psi, a_-^*(f)\Psi \rangle + 1 \leq 1, \end{aligned}$$

which implies that $\|a_-(f)\| \leq 1$, i.e. $a_-(f)$ is bounded for all $f \in \mathfrak{h}$. Similarly for $a_-^*(f)$.

An important quantity we will come back to shortly is the number density operator. Given $u \in \mathfrak{h}$ a unit vector, $\bigotimes^N u$ is an eigenvector with eigenvalue N for the operator $a^*(u)a(u)$, which means this operator “counts” how many copies of a single-particle state compose a vector in $\mathcal{F}$. This is useful when considering the occupation number basis for a bosonic or fermionic Fock space, with the important distinction that bosonic modes can have arbitrary occupation number, while fermionic modes can only have occupation 0 or 1. The last statement is the content of Pauli’s exclusion principle.

Proposition 1.8. *Let $u_1, \dots, u_r \in \mathfrak{h}$ be unit vectors and $\mathcal{H}_{n_1, \dots, n_r}$ the joint eigenspaces of the operators $a_{\pm}^*(u_1)a_{\pm}(u_1), \dots, a_{\pm}^*(u_r)a_{\pm}(u_r)$, then the fermionic and bosonic Fock spaces can be decomposed as direct sums of such eigenspaces:*

$$\begin{aligned} \mathcal{F}^B(\mathfrak{h}) &= \bigoplus_{n_1=0}^{\infty} \cdots \bigoplus_{n_r=0}^{\infty} \mathcal{H}_{n_1, \dots, n_r} \\ \mathcal{F}^F(\mathfrak{h}) &= \bigoplus_{n_1=0}^1 \cdots \bigoplus_{n_r=0}^1 \mathcal{H}_{n_1, \dots, n_r}. \end{aligned}$$

Proof. We will only prove the fermionic case, with $\dim(\mathfrak{h}) = M < \infty$ and $r = M$, as this provides the useful fermionic occupation number basis, which will be of great help to us later. Given $\{\phi_1, \dots, \phi_M\}$ an orthonormal basis for $\mathfrak{h}$, and calling $N_j = a_-^*(\phi_j)a_-(\phi_j) = a_j^*a_j$, we can prove that these number operators are commuting projections. These operators are indeed self-adjoint, as

$$N_j^* = (a_j^*a_j)^* = a_j^*a_j = N_j,$$

and idempotent, as the CAR imply

$$N_j^2 = (a_j^*a_j)^2 = -a_j^*a_j^*a_ja_j + a_j^*a_j = a_j^*a_j.$$

Moreover, for $j \neq k$, we have that

$$[N_j, N_k] = a_j^*a_ja_k^*a_k - a_k^*a_ka_j^*a_j = a_k^*a_ka_j^*a_j - a_k^*a_ka_j^*a_j = 0,$$

therefore, all the number operators are commuting projections, furthermore their spectra are all equal to $\{0, 1\}$. Note that these algebraic manipulations are well defined, since all the operators $a^\#$ (and their products) are defined on the entire space $\mathcal{F}^F(\mathfrak{h})$, which is finite-dimensional if $\mathfrak{h}$ is finite-dimensional. We now turn to the direct sum decomposition. Let $\mathbf{n} = (n_1, \dots, n_M) \in \{0, 1\}^M$ be an M -tuple of eigenvalues of the respective number operators, and define the joint projection as

$$P_{\mathbf{n}} := \prod_{i=1}^M (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)).$$

This is a product of commuting projections and is therefore still a projection, and still defined on the entire fermionic Fock space. The set $\{P_{\mathbf{n}} : \mathbf{n} \in \{0, 1\}^M\}$ is a complete set of projections, as shown by the completeness relation

$$\begin{aligned} \sum_{\mathbf{n} \in \{0, 1\}^M} P_{\mathbf{n}} &= \sum_{\mathbf{n}} \prod_{i=1}^M (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)) \\ &= \prod_{i=1}^M \sum_{n_i=0}^1 (n_i N_i + (1 - n_i)(\mathbb{1} - N_i)) \\ &= \prod_{i=1}^M \mathbb{1} = \mathbb{1}, \end{aligned}$$

again defined on the entire Fock space. We conclude by simply applying this completeness relation:

$$\mathcal{F}^F(\mathfrak{h}) = \sum_{\mathbf{n} \in \{0, 1\}^M} P_{\mathbf{n}} \mathcal{F}^F(\mathfrak{h}) = \bigoplus_{\mathbf{n} \in \{0, 1\}^M} P_{\mathbf{n}} \mathcal{F}^F(\mathfrak{h}).$$

□

Remark 12. The case where $\mathfrak{h}$ is infinite-dimensional is not much different, and even then we can always choose a finite set of M independent number operators to obtain a decomposition into a direct sum of M (possibly infinite-dimensional) joint eigenspaces. But because we will put a UV cutoff on our system the allowed momenta will form a finite set, then the one-particle Hilbert space will be finite dimensional and the above decomposition will be realized by choosing the following basis for the fermionic Fock space:

$$\{|\mathbf{n}\rangle = |n_1\rangle \otimes \dots \otimes |n_M\rangle : n_i \in \{0, 1\} \forall i = 1, \dots, M\}$$

with eigenvalues n_i given by the occupation of the M independent modes.

We can now come back to a statement that we previously used as motivation for the introduction of creation/annihilation operators, and show that indeed every observable in the grand canonical picture can be written in terms of such operators. We focus only on 1-body and 2-body observables because these are the types of operators that show up in the Hamiltonian we set out to study, but this procedure can be generalized to any N -body operator.

Theorem 1.9 (2nd quantization of 1-body operator). *Let h be an observable on $\mathfrak{h}$ and $\{u_\alpha\}_{\alpha=1}^\infty$ be an orthonormal basis for $\mathfrak{h}$ with elements in the dense domain $D(h)$. Following the notation from the beginning of this section, we can write*

$$\bigoplus_{N=1}^\infty \sum_{j=1}^N h_j = \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle_{\mathfrak{h}} a_\pm^*(u_\alpha) a_\pm(u_\beta)$$

on the domain $\cup_{M=0}^\infty \bigoplus_{N=1}^M P_\pm(D(\sum_{j=1}^N h_j))$ (for $M = 0$ the domain is taken to be $\mathbb{C}$). This operator is called the second quantization of h .

Proof. First we observe that the thesis holds in the sense of quadratic forms on $D(h)$. Indeed, let $f, g \in D(h)$ and recall the definition of creation/annihilation operators in 4

$$\begin{aligned} \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle \langle g, a_\pm^*(u_\alpha) a_\pm(u_\beta) f \rangle &= \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle \langle g, u_\alpha \rangle \langle u_\beta, f \rangle = \sum_{\beta=1}^\infty \langle g, h u_\beta \rangle \langle u_\beta, f \rangle \\ &= \sum_{\beta=1}^\infty \langle h g, u_\beta \rangle \langle u_\beta, f \rangle = \langle h g, f \rangle = \langle g, h f \rangle \end{aligned}$$

where the indices α and β are removed by using the completeness of the basis $\{u_\alpha\}_{\alpha=1}^\infty$, and h can be moved on either side of the inner products as it is self-adjoint with $f, g \in D(h)$ and $u_\alpha \in D(h)$ for all α . Moving onto N -fold pure tensors, let $f_1, \dots, f_N \in D(h)$ for some $N \geq 1$ and consider the action of $a^*(u_\alpha) a(u_\beta)$ on their tensor product

$$a^*(u_\alpha) a(u_\beta) f_1 \otimes \dots \otimes f_N = N \langle u_\beta, f_1 \rangle u_\alpha \otimes f_2 \otimes \dots \otimes f_N$$

then it follows that

$$\sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle a^*(u_\alpha) a(u_\beta) = N h_1$$

as quadratic form on finite linear combinations of pure tensor products of elements in $D(h)$. Recall the definition of the projections $P_\pm$ from Equation (3), then it is clear that

$$P_\pm N h_1 P_\pm = P_\pm \sum_{j=1}^N h_j P_\pm = \sum_{j=1}^N h_j P_\pm$$

on $\bigotimes^N \mathfrak{h}$, as each h_j acts in the same way as h on the respective factor in the tensor product, which means that it maps the (anti)-symmetric subspace into itself. Then it follows that

$$\sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle a_\pm^*(u_\alpha) a_\pm(u_\beta) P_\pm = \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle P_\pm a_\pm^*(u_\alpha) a_\pm(u_\beta) P_\pm = \bigoplus_{N=1}^\infty \sum_{j=1}^N h_j P_\pm$$

from which the thesis follows immediately. $\square$

Remark 13. Another way to lift an operator U to the Fock space is multiplicatively as $\bigoplus_{N=0}^{\infty} \bigotimes_{j=1}^N U$, and this plays an important role in the discussion of symmetries. Given a Lie group and a unitary representation γ on some Hilbert space, $d\gamma$ defines a representation of its associated Lie algebra as Hermitian operators. One can then define a map Γ that lifts the representation γ multiplicatively to the Fock space, and this induces yet another map $d\Gamma$ which lifts the representation $d\gamma$ additively to the Fock space. In general, given a unitary operator U on $\mathfrak{h}$ we may call $\Gamma(U) = \bigoplus_{N=0}^{\infty} \bigotimes_{j=1}^N U_j$ its multiplicative representation on $\mathcal{F}^{\text{F,B}}(\mathfrak{h})$, and given a self-adjoint operator A we may call $d\Gamma(A) = \bigoplus_{N=0}^{\infty} \sum_{j=1}^N A_j$ its additive representation on $\mathcal{F}^{\text{F,B}}(\mathfrak{h})$.

Remark 14. The second quantization of the identity operator on $\mathfrak{h}$ is the number operator

$$\mathcal{N} = \bigoplus_{N=0}^{\infty} \sum_{j=0}^N \mathbb{1}_j = \bigoplus_{N=0}^{\infty} N = \sum_{\alpha=1}^{\infty} a_{\pm}^*(u_{\alpha}) a_{\pm}(u_{\alpha}).$$

Now it is clear why we called $a_{\pm}^*(f) a_{\pm}(f)$ the number *density* operator, as a number operator is the sum of operators that count the occupation number of a given state over a “complete set” of states. In a much more delicate way one could define creation/annihilation *distributions* as objects that formally create/annihilate states completely localized at one point in space. If we called $|x\rangle$ an “eigenstate” (in a generalized sense that we will not discuss here) of the position operator with eigenvalue x , then the number density would be $\rho(x) = a_{\pm}^*(|x\rangle) a_{\pm}(|x\rangle)$ and the total number of particles would be written in a familiar way: $\mathcal{N} = \int_{\mathbb{R}} dx \rho(x)$, but to make sense of these objects one needs tools that go beyond the scope of our work.

Theorem 1.10 (2nd quantization of 2-body operator). *Let W be a 2-body potential for identical particles, i.e. an observable on $\mathfrak{h} \otimes \mathfrak{h}$ such that $\text{Ex}W\text{Ex} = W$. Let $\{u_{\alpha}\}_{\alpha=1}^{\infty}$ be an orthonormal basis for $\mathfrak{h}$ such that $u_{\alpha} \otimes u_{\beta} \in D(W)$ for all α and β . Then as a quadratic form on the domain $\cup_{M=2}^{\infty} \bigoplus_{N=2}^M P_{\pm}(D(\sum_{1 \leq i < j \leq N} W_{ij}))$ we have*

$$\bigoplus_{N=2}^{\infty} \sum_{1 \leq i < j \leq N} W_{ij} = \frac{1}{2} \sum_{\alpha, \beta, \nu, \mu=1}^{\infty} \langle u_{\alpha} \otimes u_{\beta}, W u_{\mu} \otimes u_{\nu} \rangle a_{\pm}^*(u_{\alpha}) a_{\pm}^*(u_{\beta}) a_{\pm}(u_{\nu}) a_{\pm}(u_{\mu}).$$

Again, this operator is called the second quantization of W . Note that in the fermionic case, the order in which the indices appear in the creation-creation and annihilation-annihilation products matters, as these operators anticommute rather than commute.

Proof. We proceed in a similar fashion to the second quantization of one-body operators. We have that

$$W u_1 \otimes u_2 = \sum_{\alpha, \beta, \mu, \nu=1}^{\infty} \langle u_{\alpha} \otimes u_{\beta}, W u_{\mu} \otimes u_{\nu} \rangle \langle u_{\mu}, u_1 \rangle \langle u_{\nu}, u_2 \rangle u_{\alpha} \otimes u_{\beta},$$

using the resolution of the identity. But now notice that the product of creation/annihilation acts on pure tensors as

$$a^*(u_{\alpha}) a^*(u_{\beta}) a(u_{\nu}) a(u_{\mu}) u_1 \otimes \cdots \otimes u_N = N(N-1) \langle u_{\mu}, u_1 \rangle \langle u_{\nu}, u_2 \rangle u_{\alpha} \otimes u_{\beta} \otimes u_3 \otimes \cdots \otimes u_N,$$

and that, like in the one-body case, the projectors $P_{\pm}$ act on W as

$$\frac{N(N-1)}{2} P_{\pm} W_{12} P_{\pm} = \sum_{1 \leq i < j \leq N} W_{ij} P_{\pm},$$

where we used that $\text{Ex}W\text{Ex} = W$. Recall, finally, that $P_{\pm} a^*(f) P_{\pm} = P_{\pm} a^*(f)$ as it doesn't matter if the starting state is (anti-)symmetrized since the image gets (anti-)symmetrized. With this in

mind, we may write

$$\begin{aligned}
\bigoplus_{N=2}^{\infty} \sum_{1 \leq i < j \leq N} W_{ij} P_{\pm} &= \bigoplus_{N=2}^{\infty} \frac{N(N-1)}{2} P_{\pm} W_{12} P_{\pm} \\
&= P_{\pm} \frac{1}{2} \sum_{\alpha, \beta, \nu, \mu=1}^{\infty} \langle u_{\alpha} \otimes u_{\beta}, W u_{\mu} \otimes u_{\nu} \rangle a^*(u_{\alpha}) a^*(u_{\beta}) a(u_{\nu}) a(u_{\mu}) P_{\pm} \\
&= \frac{1}{2} \sum_{\alpha, \beta, \nu, \mu=1}^{\infty} \langle u_{\alpha} \otimes u_{\beta}, W u_{\mu} \otimes u_{\nu} \rangle P_{\pm} a^*(u_{\alpha}) P_{\pm} a^*(u_{\beta}) a(u_{\nu}) a(u_{\mu}) P_{\pm} \\
&= \frac{1}{2} \sum_{\alpha, \beta, \nu, \mu=1}^{\infty} \langle u_{\alpha} \otimes u_{\beta}, W u_{\mu} \otimes u_{\nu} \rangle a_{\pm}^*(u_{\alpha}) a_{\pm}^*(u_{\beta}) a_{\pm}(u_{\nu}) a_{\pm}(u_{\mu}) P_{\pm},
\end{aligned}$$

which concludes the proof. $\square$

1.3 One-particle density matrices

It is clear from our discussion of the second quantization of an operator, that expectation values of any observable will inevitably reduce to expectation values of products of creation and annihilation operators. An important result that we will discuss in the next section states that – under appropriate conditions which are not at all obvious – the expectation of an observable can be written in terms of expectations of operators of the form $a_{\pm}^{\#} a_{\pm}^{\#}$. We therefore begin the study of these kinds of expectations, starting from the following definition.

Definition 1.15 (One-particle density matrix (1-pdm)). Let $\Psi = \bigoplus_{N=0}^{\infty} \Psi_N$ be a normalized vector on the bosonic or fermionic Fock space $\mathcal{F}^{\text{B,F}}(\mathfrak{h})$ with finite particle number expectation

$$\langle \Psi, \mathcal{N} \Psi \rangle = \sum_{N=0}^{\infty} N \|\Psi_N\|^2 < \infty.$$

We define the *1-particle density matrix* or *2-point function* of Ψ as the operator γ_{Ψ} on $\mathfrak{h}$ given, as a quadratic form, by

$$\langle f, \gamma_{\Psi} g \rangle_{\mathfrak{h}} = \langle \Psi, a_{\pm}^*(g) a_{\pm}(f) \Psi \rangle$$

for all $f, g \in \mathfrak{h}$.

Proposition 1.11. *The operator γ_{Ψ} is positive semi-definite, i.e. $\gamma_{\Psi} \geq 0$, and trace class, with*

$$\text{Tr}(\gamma_{\Psi}) = \langle \Psi, \mathcal{N} \Psi \rangle.$$

Proof. ??? $\square$

Lemma 1.12. *If Ψ is a finite linear combination of pure tensor products of elements from a subspace $X \subseteq \mathfrak{h}$ then γ_{Ψ} is a finite rank operator whose range is a subspace of X .*

Proof. ??? $\square$

Lemma 1.13. *If $\Psi \in \mathcal{F}^{\text{F}}(\mathfrak{h})$ is a normalized vector then γ_{Ψ} satisfies the inequality*

$$0 \leq \gamma_{\Psi} \leq \mathbb{1}.$$

In particular, the eigenvalues of γ_{Ψ} lie in the interval $[0, 1]$.

Proof. We need to show that for every $f \in \mathfrak{h}$ it holds that $0 \leq \langle f, \gamma_\Psi f \rangle \leq \|f\|^2$. Using that $a(f)$ and $a^*(f)$ are adjoints of each other, we can see that

$$\langle f, \gamma_\Psi f \rangle = \langle \Psi, a_-^*(f) a_-(f) \Psi \rangle = \langle a_-(f) \Psi, a_-(f) \Psi \rangle = \|a_-(f) \Psi\|^2 \geq 0.$$

On the other hand, the CAR imply

$$\begin{aligned} \langle f, \gamma_\Psi f \rangle &= \langle \Psi, a_-^*(f) a_-(f) \Psi \rangle \leq \langle \Psi, a_-^*(f) a_-(f) \Psi \rangle + \langle a_-^*(f) \Psi, a_-^*(f) \Psi \rangle \\ &= \langle \Psi, (a_-^*(f) a_-(f) + a_-(f) a_-^*(f)) \Psi \rangle = \langle \Psi, \{a_-^*(f), a_-(f)\} \Psi \rangle = \|f\|^2. \end{aligned}$$

This concludes the proof. $\square$

Lemma 1.14. *If $u_1, \dots, u_N$ are orthonormal vectors in $\mathfrak{h}$, then the 1pdm of $\Psi = u_1 \wedge \dots \wedge u_N$ is the projection onto the N -dimensional space $\text{span}\{u_1, \dots, u_N\}$.*

Proof. ??? $\square$

We are now ready to prove a fundamental theorem on which the treatment of any fermionic system is built, especially if perturbation theory is involved: the ground state energy and eigenvector of non-interacting fermions.

Theorem 1.15 (Ground state of non-interacting fermions). *Consider the Hamiltonian $H_N^{\text{in}} = \sum_{j=1}^N h_j$ for N particles, introduced in Equation 1, restricted to the anti-symmetric subspace $\bigwedge^N \mathfrak{h}$, i.e. with domain*

$$D_-(H_N^{\text{in}}) = P_- D(H_N^{\text{in}}).$$

Assume that the one-body operator h is bounded below, i.e. there exists some constant $E_0 \in \mathbb{R}$ such that $h \geq E_0$. The ground state energy of this fermionic system is

$$\inf \{ \text{Tr}(Ph) : P \text{ is a projection onto an } N\text{-dimensional subspace of } D(h) \}.$$

If the infimum is attained for an N -dimensional projection P then H_N^{in} has a ground state eigenvector $f_1 \wedge \dots \wedge f_N$, where $f_1, \dots, f_N$ is an orthonormal basis for $P(\mathfrak{h})$. In particular, these vectors can be chosen to be eigenvectors of h , and the expectations of h restricted to $P(\mathfrak{h})^\perp \cap D(h)$ will be greater than the expectations of h restricted to $P(\mathfrak{h})$.

Proof. Since $D(h)$ is assumed to be dense, we can choose an orthonormal basis $\{u_\alpha\}_{\alpha=1}^\infty$ for $\mathfrak{h}$ of elements in $D(h)$. Let $\Psi \in D_-(H_N^{\text{in}})$ be a normalized vector, then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{\alpha, \beta=1}^\infty \langle u_\alpha, h u_\beta \rangle \langle \Psi, a_-^*(u_\alpha) a_-(u_\beta) \Psi \rangle = \sum_{\alpha, \beta=1}^\infty h_{\alpha\beta} \langle u_\beta, \gamma_\Psi u_\alpha \rangle.$$

Since, from the second quantization of an observable, Ψ is assumed to be a finite linear combination of pure tensor products of elements in $D(h)$, it follows from Lemma 1.12 that γ_Ψ has finite rank. Let $\{v_1, \dots, v_n\}$ be the set of eigenvectors of γ_Ψ with eigenvalues $\{\lambda_1, \dots, \lambda_n\}$, then it follows from the same lemma that $v_1, \dots, v_n \in D(h)$. Then

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle = \sum_{j=1}^n \sum_{\alpha, \beta=1}^\infty \lambda_j h_{\alpha\beta} \langle u_\beta, v_j \rangle \langle v_j, u_\alpha \rangle = \sum_{j=1}^n \lambda_j \langle v_j, h v_j \rangle.$$

Because $\Psi \in D_-(H_N^{\text{in}})$, it must have fixed particle number N , then it follows that

$$\sum_{j=1}^n \lambda_j = \text{Tr}(\gamma_\Psi) = \langle \Psi, \mathcal{N} \Psi \rangle = N,$$

but $0 \leq \lambda_j \leq 1$, and therefore it must be that $n \geq N$. We may assume the eigenvectors v_j to be ordered such that $\langle v_1, hv_1 \rangle \leq \dots \leq \langle v_n, hv_n \rangle$ so that, by defining P as the projection onto $\text{span}\{v_1, \dots, v_n\}$, we can provide the bound

$$\text{Tr}(Ph) = \sum_{j=1}^N \langle v_j, hv_j \rangle \leq \sum_{j=1}^n \lambda_j \langle v_j, hv_j \rangle = \langle \Psi, H_N^{\text{in}} \Psi \rangle.$$

Thus

$$\begin{aligned} \inf\{\text{Tr}(Ph) : P \text{ proj. onto } N\text{-dimensional subspace of } D(h)\} \\ \leq \inf\{\langle \Psi, H_N^{\text{in}} \Psi \rangle : \Psi \in D(H_N^{\text{in}}), \|\Psi\| = 1\}. \end{aligned}$$

It remains to show the opposite inequality. Let $\Psi = f_1 \wedge \dots \wedge f_N$ with $f_1, \dots, f_N \in D(h)$ orthonormal vectors. Again $\gamma_\Psi =: P$ is a projection onto $\text{span}\{f_1, \dots, f_N\}$, and as above we find that

$$\langle \Psi, H_N^{\text{in}} \Psi \rangle \leq \text{Tr}(\gamma_\Psi h)$$

and the inequality carries to the infimum, which finally proves the double inequality and the first part of the thesis. Assume now that P is a minimizing projection for the variational problem. It is clear that the state $\Psi = f_1 \wedge \dots \wedge f_N$ whose 1pdm coincides with P is a ground state for H_N^{in} if $f_1, \dots, f_N$ form an orthonormal basis for $P(\mathfrak{h})$.

We may now prove the stated properties for this minimizing projection. Let $\phi \in P(\mathfrak{h})$ and $\psi \in P(\mathfrak{h})^\perp \cap D(h)$ be two normalized states, then

$$\langle \phi, h\phi \rangle \leq \langle \psi, h\psi \rangle.$$

Furthermore, consider the projection Q onto the N -dimensional space where ψ and ϕ are swapped, i.e. onto

$$\text{span}(\{\psi\} \cup (P(\mathfrak{h}) \cap \{\phi\}^\perp)),$$

and since P is a minimizer it is clear that

$$0 \leq \text{Tr}(Qh) - \text{Tr}(Ph) = \langle \psi, h\psi \rangle - \langle \phi, h\phi \rangle.$$

We can use this same argument to show that if $\psi \in D(h) \cap (P(\mathfrak{h}) \cap \{\phi\}^\perp)^\perp$ then $\langle \phi, h\phi \rangle \leq \langle \psi, h\psi \rangle$, and we now use this fact to show that h maps $P(\mathfrak{h})$ which leads to the conclusion that $P(\mathfrak{h})$ is indeed spanned by eigenvectors of h .

We proceed by contradiction, so assume that there exists a vector $\phi \in P(\mathfrak{h})$ such that $h\phi \notin P(\mathfrak{h})$. Since $D(h)$ is dense there is a $g \in D(h)$ such that $\langle (\mathbb{1} - P)g, h\phi \rangle = \langle g, (\mathbb{1} - P)h\phi \rangle \neq 0$, and, without loss of generality, we may assume $\text{Re}\langle (\mathbb{1} - P)g, h\phi \rangle \neq 0$. We have that $(\mathbb{1} - P)g \in D(h)$. Consider $t \in \mathbb{R}$ (close to 0) and define

$$\phi_t = \frac{\phi + t(\mathbb{1} - P)g}{\|\phi + t(\mathbb{1} - P)g\|}.$$

Note that $\phi_t \in D(h) \cap (P(\mathfrak{h}) \cap \{\phi\}^\perp)^\perp$, and therefore $\langle \phi, h\phi \rangle \leq \langle \phi_t, h\phi_t \rangle$. But this is a contradiction, since

$$0 = \frac{d}{dt} \Big|_{t=0} \langle \phi_t, h\phi_t \rangle = \langle (\mathbb{1} - P)g, h\phi \rangle + \langle \phi, h(\mathbb{1} - P)g \rangle = 2\text{Re}\langle (\mathbb{1} - P)g, h\phi \rangle \neq 0.$$

□

Before going over the next section there is an important generalization to be made. Everything up to now relied on the possibility of writing the second quantization of any observable by means of products of creation and annihilation operators, once a basis for the one-particle Hilbert space was chosen. It is, however, quite natural to ask what happens - among other things - when a different basis is chosen, or when the Hamiltonian contains terms of the form $a^*(f)a^*(g)$ and $a(f)a(g)$. These, and other questions, can be answered by appropriately generalizing the one-particle density matrix, and the aim of this section is precisely to introduce this generalization.

Definition 1.16 (Generalized 1pdm). Given a normalized vector $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ with finite particle number expectation, i.e. $\langle \Psi, \mathcal{N}\Psi \rangle < \infty$, we define the *generalized one-particle density matrix* as the positive semi-definite operator $\Gamma_\Psi : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ defined by

$$\langle f_1 \oplus Jg_1, \Gamma_\Psi f_2 \oplus Jg_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} = \langle \Psi, (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1))\Psi \rangle_{\mathcal{F}^{\text{F,B}}}$$

where J is the canonical isomorphism between $\mathfrak{h}$ and $\mathfrak{h}^*$ (see Appendix (A.1)).

Given the normalized vector Ψ we may write Γ_Ψ in block-matrix form by using an auxiliary operator, $\alpha_\Psi : \mathfrak{h}^* \rightarrow \mathfrak{h}$, that “measures” how far the state Ψ is from having a fixed particle number. The operator is defined by

$$\langle f, \alpha_\Psi Jg \rangle = \langle \Psi, a_\pm(g)a_\pm(f)\Psi \rangle$$

for all $f, g \in \mathfrak{h}$. Note that this map has a useful property, which essentially relates its adjoint to the corresponding operator $a_\pm^*(f)a_\pm^*(g)$, that reads

$$\alpha_\Psi^* = \pm J\alpha_\Psi J,$$

as can be shown by noting that, using the CAR or the CCR,

$$\langle \alpha_\Psi^* f, Jg \rangle = \langle f, \alpha_\Psi Jg \rangle = \pm \langle g, \alpha_\Psi Jf \rangle = \pm \langle J\alpha_\Psi Jf, Jg \rangle.$$

With this operator we come to the following lemma.

Lemma 1.16. *Given a normalized state $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ with finite particle number expectation, its generalized one-particle density matrix can be written in the block-matrix form as*

$$\Gamma_\Psi = \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix}$$

Proof. It suffices to prove that, as a quadratic form on $\mathfrak{h} \oplus \mathfrak{h}^*$, this matrix reproduces exactly Definition (1.16). Let $f_1, g_1, f_2, g_2 \in \mathfrak{h}$, then it holds that

$$\begin{aligned} (f_1 \quad Jg_1) \begin{pmatrix} \gamma_\Psi & \alpha_\Psi \\ \pm J\alpha_\Psi J & \mathbb{1} \pm J\gamma_\Psi J^* \end{pmatrix} \begin{pmatrix} f_2 \\ Jg_2 \end{pmatrix} &= (f_1 \quad Jg_1) \begin{pmatrix} \gamma_\Psi f_2 + \alpha_\Psi Jg_2 \\ \pm J\alpha_\Psi Jf_2 + (\mathbb{1} \pm J\gamma_\Psi J^*)Jg_2 \end{pmatrix} \\ &= \langle f_1, \gamma_\Psi f_2 \rangle + \langle f_1, \alpha_\Psi Jg_2 \rangle + \langle Jg_1, \alpha_\Psi^* f_2 \rangle + \langle Jg_1, (\mathbb{1} \pm J\gamma_\Psi J^*)Jg_2 \rangle \\ &= \langle a_\pm^*(f_2)a_\pm(f_1) \rangle + \langle a_\pm(g_2)a_\pm(f_1) \rangle + \langle a_\pm^*(f_2)a_\pm^*(g_1) \rangle + \langle a_\pm(g_2)a_\pm^*(g_1) \rangle \\ &= \langle (a_\pm^*(f_2) + a_\pm(g_2))(a_\pm(f_1) + a_\pm^*(g_1)) \rangle \\ &= \langle f_1 \oplus Jg_1, \Gamma_\Psi f_2 \oplus Jg_2 \rangle. \end{aligned}$$

□

The usefulness of this new object is that it allows for a very simple way to write this quantity in a way that closely resembles the standard one-particle density matrix. We just need to define the generalized creation/annihilation operators:

Definition 1.17. Let $f, g \in \mathfrak{h}$ be one-particle states. The generalized annihilation and creation operators are defined, respectively, by the maps

$$\begin{aligned} A_{\pm}(f \oplus Jg) &:= a_{\pm}(f) + a_{\pm}^*(g) \\ A_{\pm}^*(f \oplus Jg) &:= a_{\pm}^*(f) + a_{\pm}(g). \end{aligned}$$

Remark 15. Once again we have that the generalized annihilation operators are conjugate linear from $\mathfrak{h} \oplus \mathfrak{h}^*$ to operators on $\mathcal{F}^{F,B}(\mathfrak{h})$. Furthermore, we have the following relation between these generalized operators. For all $F \in \mathfrak{h} \oplus \mathfrak{h}^*$ it holds that

$$A_{\pm}^*(F) = A_{\pm}(\mathcal{J}F), \quad \text{where } \mathcal{J} = \begin{pmatrix} 0 & J^* \\ J & 0 \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$$

Proposition 1.17 (Generalized CCR and CAR). *For all $F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$ we have the following generalized versions of the canonical (anti-)commutation relations. For bosons*

$$[A_+(F_1), A_+^*(F_2)] = \langle F_1, \mathcal{S}F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} \quad \text{where } \mathcal{S} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*,$$

and for fermions

$$\{A_-(F_1), A_-^*(F_2)\} = \langle F_1, F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*}.$$

More compactly we could introduce the natural notation $[A_{\pm}(F_1), A_{\pm}^(F_2)]_{\mp} = \langle F_1, \mathcal{S}_{\pm}F_2 \rangle$, where we define $\mathcal{S}_{-}$ as simply being the identity, and the sign on the square brackets indicates a commutator (-) or an anticommutator (+).*

Proof. The proof is trivial, as it only requires to expand the expression for the (anti-)commutator and to use explicitly the CCR/CAR. $\square$

Remark 16. Note that, in general, the (anti-)commutators $[A_{\pm}(F), A_{\pm}(G)]_{\mp}$ and $[A_{\pm}^*(F), A_{\pm}^*(G)]_{\mp}$ need not be zero.

The special case where $F = f \oplus 0$ reduces to the usual creation/annihilation operators because $A_{\pm}^{\#}(f \oplus 0) = a_{\pm}^{\#}(f)$, which means that this is by all means an extension of the theory we developed so far. With these operators, the generalized one-particle density matrix can be written in a familiar form. Given, as usual, $\Psi \in \mathcal{F}^{F,B}(\mathfrak{h})$ normalized and for all $F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$ we have

$$\langle F_1, \Gamma_{\Psi} F_2 \rangle_{\mathfrak{h} \oplus \mathfrak{h}^*} = \langle \Psi, A_{\pm}^*(F_1) A_{\pm}(F_2) \Psi \rangle$$

1.4 Bogolubov transformations and quasi-free pure states

We can finally realize the notion of a “change of modes” for creation/annihilation operators on the space $\mathfrak{h} \oplus \mathfrak{h}^*$. The following definition characterizes a class of isomorphisms on this space as those that, in some sense, preserve the structure of the generalized creation and annihilation operators.

Definition 1.18 (Bogolubov transformations). A linear bounded isomorphism $\mathcal{V} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ is called a (bosonic or fermionic) *Bogolubov map* if, for all $F, G \in \mathfrak{h} \oplus \mathfrak{h}^*$

- $A^*(\mathcal{V}F) = A(\mathcal{V}\mathcal{J}F)$;
- $[A_{\pm}(\mathcal{V}F), A_{\pm}^*(\mathcal{V}G)]_{\mp} = \langle F, \mathcal{S}_{\pm}G \rangle$.

One also refers to the map $A_{\pm}(F) \mapsto A_{\pm}(\mathcal{V}F)$ as a *Bogolubov transformation*.

We can characterize Bogolubov maps by writing them in block-matrix form and noting that each block must satisfy specific properties that come directly from the definition. We first notice that the conditions in the definition can be rewritten as

$$\langle \mathcal{V}F_1, \mathcal{S}_\pm \mathcal{V}F_2 \rangle = \langle F_1, \mathcal{S}_\pm F_2 \rangle \quad \text{and} \quad \mathcal{J}\mathcal{V}F = \mathcal{V}\mathcal{J}F$$

for all $F, F_1, F_2 \in \mathfrak{h} \oplus \mathfrak{h}^*$. Moreover, since we are assuming that $\mathcal{V}$ is invertible, we can also write

$$\mathcal{V}^{-1} = \mathcal{S}_\pm \mathcal{V}^* \mathcal{S}_\pm.$$

In particular, $\mathcal{V}^{-1} = \mathcal{V}^*$ in the fermionic case. With these properties we immediatly get to the following proposition, for which the proof can be read through what we just said:

Proposition 1.18. *A linear map $\mathcal{V}$ on $\mathfrak{h} \oplus \mathfrak{h}^*$ is a Bogolubov map if, and only if, the following conditions hold:*

$$\mathcal{V}^* \mathcal{S}_\pm \mathcal{V} = \mathcal{S}_\pm, \quad \mathcal{V} \mathcal{S}_\pm \mathcal{V}^* = \mathcal{S}_\pm, \quad \mathcal{J}\mathcal{V}\mathcal{J} = \mathcal{V}. \quad (5)$$

If we take bounded linear maps $U : \mathfrak{h} \rightarrow \mathfrak{h}$ and $V : \mathfrak{h} \rightarrow \mathfrak{h}^*$ we can finally write a Bogolubov map in block-matrix form

$$\mathcal{V} = \begin{pmatrix} U & J^* V J^* \\ V & J U J^* \end{pmatrix} \quad (6)$$

and the conditions (5) can then be read as

$$\begin{cases} U^* U = \mathbb{1} \pm V^* V \\ U U^* = \mathbb{1} \pm J^* V^* J \\ J V^* J U = \pm J U^* J^* V \end{cases}.$$

Defining Bogolubov maps is not sufficient though, as such a map need not, in general, correspond to a unitary on the Fock space. Thus we come to the following theorem which tells us when this is possible, and which we will use to define an important class of states in a Fock space for which Wick's theorem will hold in its most general form.

Theorem 1.19 (Unitary implementability of Bogolubov maps). *If $\mathcal{V} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ is a Bogolubov map of the form (6) then there exists a unitary map*

$$\mathbb{U}_\mathcal{V} : \mathcal{F}^{\mathbb{F}, \mathbb{B}}(\mathfrak{h}) \rightarrow \mathcal{F}^{\mathbb{F}, \mathbb{B}}(\mathfrak{h})$$

such that, for $F \in \mathfrak{h} \oplus \mathfrak{h}^*$,

$$\mathbb{U}_\mathcal{V} A_\pm(F) \mathbb{U}_\mathcal{V}^* = A_\pm(\mathcal{V}F)$$

if, and only if, $V^* V$ is trace-class. This trace-class condition is also referred to as the Shale-Stinespring condition.

What this means is that, under suitable conditions on the block-matrix form of a Bogolubov map, there exists a “change of basis” on the Fock space that sends generalized annihilation and - by taking the adjoint - creation operators to operators that annihilate/create a superposition of states determined by the linear map $\mathcal{V}$.

Proof. AAAAAAHHHHHHHHH □

We are finally ready to discuss the most important result of this chapter: Wick's theorem. This is what, among other things, allows us to write perturbative series in the way described in the next chapter, and it motivates tools we will need to introduce. The content of this important theorem is, concretely, that all the expectations we are interested in reduce to products of the 2-point function. We first look at the case of pure states and then move onto Gibbs states.

Definition 1.19 (Quasi-free pure state). A vector $\Psi \in \mathcal{F}^{\text{F,B}}(\mathfrak{h})$ is called a *quasi-free pure state* if there exists a unitarily implementable Bogolubov map $\mathcal{V}$ such that $\Psi = \mathbb{U}_{\mathcal{V}}|0\rangle$, where $\mathbb{U}_{\mathcal{V}}$ is the unitary implementation of $\mathcal{V}$.

That is, a pure state is quasi-free if it is the vacuum state under an appropriate choice of physical modes. **DEFINE CONTRACTIONS TO PROVE WICK'S THEOREM.**

Theorem 1.20 (Wick's theorem). *If Ψ is a quasi-free pure state and $F_1, \dots, F_{2m} \in \mathfrak{h} \oplus \mathfrak{h}^*$ for some $m \geq 1$, then it holds that*

$$\begin{aligned} \langle \Psi, A_{\pm}(F_1) \cdots A_{\pm}(F_{2m}) \Psi \rangle = \\ \sum_{\sigma \in P_{2m}} (\pm 1)^{\sigma} \langle \Psi, A_{\pm}(F_{\sigma(1)}) A_{\pm}(F_{\sigma(2)}) \Psi \rangle \cdots \langle \Psi, A_{\pm}(F_{\sigma(2m-1)}) A_{\pm}(F_{\sigma(2m)}) \Psi \rangle \end{aligned} \quad (7)$$

and

$$\langle \Psi, A_{\pm}(F_1) \cdots A_{\pm}(F_{2m-1}) \Psi \rangle = 0$$

where P_{2m} is the set of pairings of the indices $\{1, \dots, 2m\}$, i.e.

$$P_{2m} = \{ \sigma \in S_{2m} : \sigma(2j-1) < \sigma(2j+1), j = 1, \dots, m-1 \text{ and } \sigma(2j-1) < \sigma(2j), j = 1, \dots, m \}.$$

Proof. The theorem is trivially true for the case $m = 1$, and we will prove the case $m > 1$ by induction. $\square$

We may now extend the definition of quasi-free pure states to include non-pure ones, and we may as well use Wick's theorem as the defining property of quasi-free expectations. Surprisingly enough, Gibbs states are quasi-free, as illustrated by the theorem we are about to prove.

Definition 1.20 (General quasi-free states). A *quasi-free state* is a quantum state $\rho \in \mathfrak{B}^*(\mathcal{F}^{\text{F,B}}(\mathfrak{h}))$ that satisfies Wick's theorem, i.e. such that Equation (7) holds when replacing expectations $\langle \Psi, \cdot \Psi \rangle$ with mixed expectations $\langle \cdot \rangle_{\rho}$.

Theorem 1.21. *Let $\{F_n\}_{n \in \mathbb{N}}$ be an orthonormal basis for $\mathfrak{h} \oplus \mathfrak{h}^*$ and define the Hamiltonian $H = \sum_{n=0}^{\infty} \epsilon_n A_{\pm}^*(F_n) A_{\pm}(F_n)$, where $\text{Tr}(e^{-\beta H}) = \sum_n e^{-\beta \epsilon_n} < \infty$. The Gibbs state $\langle \cdot \rangle_{\beta, H}$ is quasi-free.*

Proof. **LOOK AT GIULIANI AND VIGNALE.** $\square$

1.5 Quadratic hamiltonians

Having introduced all the necessary results regarding Bogolubov maps, we can now move onto the study of generic quadratic Hamiltonians. Their importance is partly due to the fact that if one can find a Bogolubov transformation that diagonalizes the quadratic part of the Hamiltonian, then the study of the kinetic (quadratic) part of that system is automatically done. This of course plays a crucial role in perturbative calculations, which are all about dealing with complicated interactions on top of an easy and well-defined “free theory”, i.e. a theory whose n -point functions satisfy Wick's theorem.

Definition 1.21 (Quadratic Hamiltonian). Let $\mathcal{A} : \mathfrak{h} \oplus \mathfrak{h}^* \rightarrow \mathfrak{h} \oplus \mathfrak{h}^*$ be a self-adjoint, trace-class operator and assume moreover in the bosonic case that it is positive semi-definite. Given an orthonormal basis $\{F_1, \dots, F_{2M}\}$ for $\mathfrak{h} \oplus \mathfrak{h}^*$, the operator

$$H_{\mathcal{A}}^{\pm} = \sum_{i,j=1}^{2M} \langle F_i, \mathcal{A} F_j \rangle A_{\pm}^*(F_i) A_{\pm}(F_j),$$

is called the *quadratic Hamiltonian associated to $\mathcal{A}$* .

2 Perturbation Theory

In the previous chapter we built the second quantization formalism to be able to introduce, in full generality, the key property that allows to greatly simplify the computation of expectation values of any observable. Looking at the Gibbs state functional

$$\langle \cdot \rangle_\beta = \frac{\text{Tr}(\cdot e^{-\beta H})}{\text{Tr}(e^{-\beta H})}$$

we can imagine expanding the exponential in the numerator as (something similar to) a power series, and consider each term as a contribution at a specific order. Such terms will contain powers of the energy operators, i.e. polynomials of creation/annihilation operators. Then Wick's theorem tells us that at all orders we can write the correct contribution using more and more complicated combinations of the 2-point function. These combinations can be easily found by means of the very clever diagrammatic approach first employed by Richard Feynman, and the aim of this chapter is to outline the inner workings of this approach in the specific case of a fermionic theory.

Remark 17. We will only deal with the perturbative series for thermal equilibrium states, but the idea behind the tools we will introduce shortly is the same as the one used for scattering processes in any QFT. In fact, there is a beautiful connection between the calculations done for such processes and the theory we build below, and it has to do with the link between the time evolution operator of quantum mechanics e^{-iHt} and its imaginary time version $e^{-\tau H}$, which looks exactly like what appears in a Gibbs state.

2.1 A toy model: bosonic expectations

We build the theory of Feynman diagrams, or graphs, by first looking at a simple model: a classical Hamiltonian given by polynomials of complex variables. We will introduce expectations as Gaussian integrals over such variables and show that they exactly reproduce Wick's theorem, in a sense that will be made clear in the next section. After that, we will interpret this tool as a formal way of writing expectations of bosonic (i.e. *commuting*) operators, and then introduce an algebraic structure capable of carrying the results found to the fermionic (i.e. *anticommuting*) case.

Suppose we are given a classical system of N coupled harmonic oscillators: a linear chain model, with Hamiltonian

$$H_0 = \sum_{j \in \mathbb{Z}_N} \left(\frac{p_j^2}{2} + (q_{j+1} - q_j)^2 \right).$$

It can be shown that there exists a canonical transformation that sends H_0 into

$$H_0 = \sum_{k \in \mathcal{D}} \omega_k z_k^* z_k$$

where $\omega_k = 2 \sin(\frac{\pi k}{N})$ and $\mathcal{D} = \{k \in \mathbb{Z} : -\frac{N}{2} \leq k \leq \frac{N+1}{2}\}$. Following the classical theory, the partition function for this system is

$$Z_0(\beta) \propto \int \left(\prod_{k \in \mathcal{D}} dz_k^* dz_k \right) e^{-\beta H_0} = \prod_{k \in \mathcal{D}} \int dz_k^* dz_k e^{-\beta \omega_k z_k^* z_k} \propto \prod_{k \in \mathcal{D}} \frac{1}{\beta \omega_k}$$

where, recall, the integral is given by

$$\int dz^* dz f(z, z^*) := 2i \int_{\mathbb{R}^2} dx dy f(x + iy, x - iy).$$

Suppose now we want to add anharmonic corrections to the model. We start with a quartic term, and already we notice that exact diagonalization is not possible anymore. Performing the canonical transformation into the variables z_k and z_k^* leads to

$$\begin{aligned} H = H_0 + V &= \sum_{j \in \mathbb{Z}_N} \left(\frac{p_j^2}{2} + (q_{j+1} - q_j)^2 + \frac{\lambda}{4} (q_{j+1} - q_j)^4 \right) \\ &= \sum_{k \in \mathcal{D}} \omega_k z_k^* z_k + \lambda \sum_{k \in \mathcal{D}} \Omega_{p,q,k} z_{q+k}^* z_{p-k}^* z_p z_q. \end{aligned}$$

Now the partition function reads as

$$Z(\beta, \lambda) = \int \left(\prod_{k \in \mathcal{D}} dz_k^* dz_k \right) e^{-\beta H_0} e^{-\beta \lambda V}, \quad (8)$$

and this is not exactly computable. If we suppose that $\lambda \ll 1$, we can expand the exponential of the correction term as a power series and truncate $Z(\beta, \lambda)$ to the desired order. For example

$$\begin{aligned} Z(\beta, \lambda) &= \int \left(\prod_{k \in \mathcal{D}} dz_k^* dz_k \right) e^{-\beta H_0} \left(1 - \lambda \beta V + \mathcal{O}(\lambda^2) \right) \\ &= Z_0(\beta) \left(1 - \lambda \beta \frac{\int \left(\prod_{k \in \mathcal{D}} dz_k^* dz_k \right) e^{-\beta H_0} V}{Z_0(\beta)} + \mathcal{O}(\lambda^2) \right). \end{aligned}$$

Looking at this first order correction, we could interpret the expression

$$\frac{\left(\prod_{k \in \mathcal{D}} dz_k^* dz_k \right) e^{-\beta H_0}}{Z_0(\beta)} =: P_0(dz)$$

as some Gaussian probability measure, and correspondingly interpret $\int P_0(dz) V$ as a kind of average of V . Higher order corrections can also be interpreted as higher moments of V with respect to this Gaussian probability measure.

We now want to show that this kind of expectation satisfies Wick's theorem, or Isserlis' theorem as it is known in the field of probability theory. After that it will be clear that the variables z_k^* and z_k may just as well be regarded to as a formal representation of creation/annihilation operators, and higher order corrections to the free partition function $Z_0(\beta)$ are going to be expectations of powers of the interaction potential V . The only obstacle is then the fact that complex Gaussian integrals actually satisfy the *bosonic* version of Wick's theorem, and this is a consequence of z_k^* and z_k being commuting variables rather than anticommuting. The missing fermionic signs will be the subject of Section 2.2.

We start our discussion of bosonic expectations by recalling some properties of complex Gaussian integrals.

Definition 2.1 (Complex covariance). A matrix $G \in \text{Mat}_{N \times N}$ is said to be a *complex covariance* if

- (i) there exists a matrix U that diagonalizes G ;
- (ii) all eigenvalues of G have positive real part.

The space of complex covariances on will be denoted by $\mathcal{C}_N$.

To ease the notation, if $\mathbf{z}, \mathbf{w} \in \mathbb{C}^N$ we will write $\mathbf{z} \cdot \mathbf{w} = \sum_{j=1}^N z_j w_j$.

Lemma 2.1. *Let $G \in \mathcal{C}_N$ and $\mathbf{z} \in \mathbb{C}^N$, then*

$$\int \left(\prod_{j=1}^N \frac{dz_j^* dz_j}{2\pi i} \right) e^{-\mathbf{z} \cdot G^{-1} \mathbf{z}} = \det(G).$$

Proof. Because $G \in \mathcal{C}_N$ there exists U that diagonalizes G , i.e. there exist ξ such that $\mathbf{z} = U\xi$. Then

$$\begin{aligned}\mathbf{z}^* \cdot G^{-1} \mathbf{z} &= \mathbf{z}^* \cdot U^\dagger G^{-1} U \mathbf{z} =: \boldsymbol{\xi}^* \cdot D^{-1} \boldsymbol{\xi} \\ d\mathbf{z}^* d\mathbf{z} &= \det(U^\dagger) \det(U) d\boldsymbol{\xi}^* d\boldsymbol{\xi} = d\boldsymbol{\xi}^* d\boldsymbol{\xi}\end{aligned}$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_N)$ is the diagonalization of G . Then, the left hand side becomes

$$\text{l.h.s.} = \prod_{j=1}^N \int \frac{d\xi_j^* d\xi_j}{2\pi i} e^{-\xi_j^* \frac{1}{\lambda_j} \xi_j} = \prod_{j=1}^N \int_{\mathbb{R}^2} \frac{dx dy}{\pi} e^{-\frac{x^2+y^2}{\lambda_j}} = \det(G).$$

□

Definition 2.2 (Complex Gaussian measure). Let $G \in \mathcal{C}_N$. The complex integration measure

$$P_G(d\mathbf{z}) := \frac{1}{\det(G)} \int \left(\prod_{j=1}^N \frac{dz_j^* dz_j}{2\pi i} \right) e^{-\mathbf{z}^* \cdot G^{-1} \mathbf{z}}$$

is called *complex Gaussian measure* of covariance G .

Remark 18. The matrix G is also called a propagator because, as we will see, it covers the role of the 2-point function, which is itself also called propagator.

Note that a complex Gaussian measure is “normalized”, in the sense that $\int P_G(d\mathbf{z}) = 1$. We are now ready to define expectation values and prove their fundamental properties.

Definition 2.3 (Complex/bosonic expectation). Given $P_G(d\mathbf{z})$ and a measurable function f , the *expectation of f* with respect to $P_G(d\mathbf{z})$ is defined as

$$\mathbb{E}_G(f) := \int P_G(d\mathbf{z}) f(\mathbf{z}, \mathbf{z}^*).$$

Remark 19. In Section 2.1, we actually had $G = \text{diag}\left(\frac{1}{\beta\omega_1}, \dots, \frac{1}{\beta\omega_N}\right)$ and the first order correction to the partition function was indeed $\mathbb{E}_G(V)$.

Lemma 2.2. Let $G \in \mathcal{C}_N$ and let $\mathbf{J} \in \mathbb{C}^N$ be a fixed complex vector, often called a *source term*. Then

$$\int P_G(d\mathbf{z}) e^{\mathbf{J}^* \cdot \mathbf{z} + \mathbf{J} \cdot \mathbf{z}^*} = e^{\mathbf{J}^* \cdot G \mathbf{J}}$$

Proof. Let U be the diagonalization matrix for G , let $\mathbf{z} = U\xi$ and $\mathbf{J} = U\eta$. Then we can rewrite the exponent of the integrand and complete the square:

$$\begin{aligned}-\mathbf{z}^* \cdot G^{-1} \mathbf{z} + \mathbf{z}^* \cdot \mathbf{J} + \mathbf{J} \cdot \mathbf{z}^* &= -\boldsymbol{\xi}^* \cdot D^{-1} \boldsymbol{\xi} + \boldsymbol{\xi}^* \cdot \boldsymbol{\eta} + \boldsymbol{\xi} \cdot \boldsymbol{\eta}^* \\ &= -(\boldsymbol{\xi}^* - D\boldsymbol{\eta}^*) \cdot D^{-1} (\boldsymbol{\xi} - D\boldsymbol{\eta}) + \boldsymbol{\eta}^* \cdot D\boldsymbol{\eta} \\ &= -(\boldsymbol{\xi}^* - D\boldsymbol{\eta}^*) \cdot D^{-1} (\boldsymbol{\xi} - D\boldsymbol{\eta}) + \mathbf{J}^* \cdot G \mathbf{J}\end{aligned}$$

where, once again, $D = \text{diag}(\lambda_1, \dots, \lambda_N)$ is the diagonalization of G . We can apply the change of variables $\xi_j \mapsto \xi_j + \frac{1}{\lambda_j} \eta_j$ and take the term $\mathbf{J}^* \cdot G \mathbf{J}$ out of the integral, but the remaining integral is just $\int P_G(d\xi) = 1$, and the thesis follows immediately. □

To prove the next theorem we will need the two following identities:

$$\begin{aligned} P_G(dz)z_j &= \frac{\partial}{\partial J_j^*} \left(P_G(dz)e^{\mathbf{J}^* \cdot \mathbf{z} + \mathbf{J} \cdot \mathbf{z}^*} \right) \Big|_{\mathbf{J}=\mathbf{J}^*=0} \\ P_G(dz)z_j^* &= \frac{\partial}{\partial J_j} \left(P_G(dz)e^{\mathbf{J}^* \cdot \mathbf{z} + \mathbf{J} \cdot \mathbf{z}^*} \right) \Big|_{\mathbf{J}=\mathbf{J}^*=0} \end{aligned} \quad (9)$$

Theorem 2.3 (Isserlis' theorem). *Given $G \in \mathcal{C}_N$, let M be a positive integer and consider two sets of indices: $\alpha_1, \dots, \alpha_M$ and $\beta_1, \dots, \beta_M$. Then it holds that*

$$\mathbb{E}_G(z_{\alpha_1} \cdots z_{\alpha_M} z_{\beta_1}^* \cdots z_{\beta_M}^*) = \sum_{\sigma \in S_N} \prod_{j=1}^M G_{\alpha_j, \beta_{\sigma(j)}}$$

where S_M is the set of permutations of M elements.

Proof. Using the identities in (9) and Lemma (2.2) we can write

$$\mathbb{E}_G(z_{\alpha_1} \cdots z_{\alpha_M} z_{\beta_1}^* \cdots z_{\beta_M}^*) = \frac{\partial^{2M}}{\partial J_{\alpha_1}^* \cdots \partial J_{\alpha_M}^* \partial J_{\beta_1} \cdots \partial J_{\beta_M}} \left(e^{\mathbf{J}^* \cdot G \mathbf{J}} \right) \Big|_{\mathbf{J}=\mathbf{J}^*=0} = \star$$

Now we can expand the exponential as a power series and derive term by term. In fact, when deriving $2M$ times the n -th term of this series we have the following cases: if $r < M$ we are taking derivatives of order higher than the degree of the polynomial so it gives zero, if $r > M$ deriving will lead to a homogeneous polynomial evaluated at zero so it gives zero, if $r = M$ some terms survive. Indeed, if we look at the derivatives with respect to the $\mathbf{J}^*$ we see that

$$\begin{aligned} \frac{\partial^M}{\partial J_{\alpha_1}^* \cdots \partial J_{\alpha_M}^*} \frac{(\mathbf{J}^* \cdot G \mathbf{J})^M}{M!} &= \frac{1}{M!} \frac{\partial^M}{\partial J_{\alpha_1}^* \cdots \partial J_{\alpha_M}^*} \left(\sum_{j,k=1}^M J_{\rho_j}^* G_{\rho_j, \tau_k} J_{\tau_k} \right)^M \\ &= \frac{\partial^{M-1}}{\partial J_{\alpha_2}^* \cdots \partial J_{\alpha_M}^*} \frac{M}{M!} \left(\sum_{j,k=1}^M \delta_{\rho_j, \alpha_1} G_{\rho_j, \tau_k} J_{\tau_k} \right) \left(\sum_{j,k=1}^M J_{\rho_j}^* G_{\rho_j, \tau_k} J_{\tau_k} \right)^{M-1} \\ &= \prod_{\ell=1}^M \sum_{j,k=1}^M \delta_{\rho_j, \alpha_\ell} G_{\alpha_\ell, \tau_k} J_{\tau_k}. \end{aligned}$$

Using this in $\star$ we get that

$$\star = \frac{\partial^M}{\partial J_{\beta_1} \cdots \partial J_{\beta_M}} \left(\prod_{j=1}^M \sum_{k=1}^M G_{\alpha_j, \tau_k} J_{\tau_k} \right) \Big|_{\mathbf{J}=\mathbf{J}^*=0}$$

and applying Leibnitz rule we arrive at the thesis. $\square$

2.2 Grassmann integrals and fermionic expectations

It is time to introduce the anticipated algebraic structure that carries the results found so far to the fermionic case: Grassmann variables. Consider the unital complex Grassmann algebra generated by a set of $2N$ anticommuting vectors:

$$\begin{aligned} \mathcal{G}_N &= \text{span}\{1, \psi_1, \bar{\psi}_1, \dots, \psi_N, \bar{\psi}_N\} \\ \{\psi_i, \psi_j\} &= 0 \quad \forall i, j = 1, \dots, N \\ \{\psi_i, \bar{\psi}_j\} &= 0 \quad \forall i, j = 1, \dots, N \\ \{\bar{\psi}_i, \bar{\psi}_j\} &= 0 \quad \forall i, j = 1, \dots, N \end{aligned} \quad (10)$$

where $\{\psi, \phi\} := \psi\phi + \phi\psi$ is the anticommutator. We need to define Gaussian integrals and expectations of Grassmann variables.

Given $\mathcal{G}_N$ the Grassmann algebra generated by $\{\psi_1, \dots, \psi_N, \bar{\psi}_1, \dots, \bar{\psi}_N\}$, denote by

$$\begin{aligned}\mathcal{G}_N^- &= \{\phi \in \mathcal{G}_N : [\phi, \psi] = 0 \ \forall \psi \in \mathcal{G}_N\} \\ \mathcal{G}_N^+ &= \{\phi \in \mathcal{G}_N : \{\phi, \psi\} = 0 \ \forall \psi \in \mathcal{G}_N\}\end{aligned}$$

the even (commuting) and odd (anticommuting) subsets of the algebra. It is easy to show that these are subspaces (by linearity of commutators and anticommutators), and one can also show that $\mathcal{G}_N$ can be decomposed as

$$\mathcal{G}_N \cong \mathcal{G}_N^+ \oplus \mathcal{G}_N^-.$$

Odd elements satisfy a really important property. If $\psi \in \mathcal{G}_N^+$ it follows that $\{\psi, \psi\} = 2\psi^2 = 0$. That is, all powers greater than one of an odd element vanish.

Definition 2.4 (Functions of Grassmann variables). Given $\mathcal{G}$ a Grassmann algebra, as defined above, and $f : \mathbb{C}^k \rightarrow \mathbb{C}$ an analytic function given by its Taylor series centered at zero

$$f(\mathbf{z}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \mathbf{z}^i$$

we define the associated *function of Grassmann variables* as

$$f(\boldsymbol{\psi}) = \sum_{j=0}^{\infty} \sum_{|\alpha|=j} \frac{\partial^\alpha f(0)}{j!} \boldsymbol{\psi}^i$$

where $\boldsymbol{\psi} \in \mathcal{G}_N \times \dots \times \mathcal{G}_N$ is the list of k Grassmann variables.

Remark 20. If $\psi \in \mathcal{G}_N^+$ it follows that any function of Grassmann variables evaluated in ψ has at most first order terms, as all higher orders will always contain a factor of $\psi^2 = 0$. The most important example to us is the exponential of an odd element:

$$e^\psi = 1 + \psi + \frac{1}{2}\psi^2 + \dots = 1 + \psi$$

We now introduce integrals of functions of Grassmann variables. Note that the definition is going to be purely algebraic and, in particular, motivated by our need to reproduce the fermionic version of Wick's theorem. We ask, first of all, for the integral to be a linear operator on a Grassmann algebra. Then, specifying the image of a basis for the algebra will uniquely determine the operator. The specific definition of the image of basis elements is such that a Grassmann integral over all such basis elements extracts the coefficient of the highest degree term of the integrand.

Definition 2.5 (Grassmann integral). Given $\{1, \psi_1, \dots, \psi_N, \bar{\psi}_1, \dots, \bar{\psi}_N\}$ the anticommuting basis for a Grassmann algebra, the *Grassmann integral* is the linear operator that acts on this basis as

$$\begin{aligned}\int d\psi_i \psi_j &= \int d\bar{\psi}_i \bar{\psi}_j := \delta_{ij}, \\ \int d\psi_i &= \int d\bar{\psi}_i := 0.\end{aligned}$$

Repeated integration is denoted by a single integral symbol and multiple “differentials”.

Example 1. In $\mathcal{G}_2$, consider the exponential function. Using the definition, we can evaluate the following integrals to see how this operator behaves.

- $\int d\psi_1 e^{\psi_1 \bar{\psi}_2} = \int d\psi_1 (1 + \psi_1 \bar{\psi}_2) = \bar{\psi}_2;$
- $\int d\bar{\psi}_2 e^{\psi_1 \bar{\psi}_2} = \int d\bar{\psi}_2 (1 - \bar{\psi}_2 \psi_1) = -\psi_1;$
- $\int d\psi_1 d\bar{\psi}_2 e^{\psi_1 \bar{\psi}_2} = -1.$

Remark 21. Repeated integration, i.e. the composition of multiple integral operators, inherits some form of anticommutativity when acting on the basis elements. Indeed in $\mathcal{G}_1$, for example, we have

$$\int d\psi d\bar{\psi} \psi \bar{\psi} = - \int d\psi d\bar{\psi} \bar{\psi} \psi = -1 = - \int d\bar{\psi} d\psi \psi \bar{\psi}. \quad (11)$$

And given the definition of Grassmann functions, this property will be quite useful to us.

Having defined integration, we can now introduce a *Gaussian measure*, again in a purely algebraic sense. To make the notation lighter, we define a “dot product”, without any geometric meaning, as $\psi \cdot \phi := \sum_{i=1}^k \psi_i \phi_i$.

Definition 2.6 (Grassmann covariance). A matrix $G \in \text{GL}(N, \mathbb{C})$ is called a *Grassmann covariance* on the Grassmann algebra $\mathcal{G}_N$.

Lemma 2.4. Let G be a Grassmann covariance, and let $\psi = (\psi_1, \dots, \psi_N)$ and $\bar{\psi} = (\bar{\psi}_1, \dots, \bar{\psi}_N)$. Then

$$\int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) e^{-\bar{\psi} \cdot G^{-1} \psi} = \det(G^{-1})$$

Proof. We carry out the calculation explicitly.

$$\begin{aligned} \int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) e^{-\bar{\psi} \cdot G^{-1} \psi} &= \int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) \frac{(-1)^N}{N!} \left(\sum_{i,j=1}^N \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\ &= \int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) \frac{1}{N!} \left(\sum_{i,j=1}^N \bar{\psi}_i (G^{-1})_{ij} \psi_j \right)^N \\ &= \int \left(\prod_{i=1}^N d\bar{\psi}_i d\psi_i \right) \frac{1}{N!} \sum_{\sigma, \tau \in S_N} \prod_{j=1}^N (G^{-1})_{\sigma(j), \tau(j)} \bar{\psi}_{\sigma(j)} \psi_{\tau(j)} = \star \end{aligned}$$

Where S_N is the set of permutations of N elements. We can split the product into a product of matrix elements times a product of Grassmann variables, then we use that

$$\prod_{j=1}^N \bar{\psi}_{\sigma(j)} \psi_{\tau(j)} = (-1)^{\text{sgn}(\sigma) + \text{sgn}(\tau)} \prod_{j=1}^N \bar{\psi}_j \psi_j$$

to find that

$$\begin{aligned} \star &= \frac{1}{N!} \sum_{\sigma, \tau \in S_N} (-1)^{\text{sgn}(\sigma) + \text{sgn}(\tau)} \prod_{j=1}^N (G^{-1})_{\sigma(j), \tau(j)} \\ &= \frac{1}{N!} \sum_{\sigma \in S_N} (-1)^{\text{sgn}(\sigma)} \prod_{j=1}^N (G^{-1})_{j, \sigma j} \\ &= \det(G^{-1}) \end{aligned}$$

□

Definition 2.7 (Grassmann Gaussian measure). Let G be a Gaussian covariance. A *Grassmann Gaussian measure* is the Grassmann integral measure defined by

$$P_G(d\psi) := \frac{1}{\det(G^{-1})} \left(\prod_{j=1}^N d\bar{\psi}_j d\psi_j \right) e^{-\bar{\psi} \cdot G^{-1} \psi}$$

where ψ_i and $\bar{\psi}_i$ are the usual anticommuting Grassmann variables.

Remark 22. The factor in front of the differentials guarantees a “normalized” measure, i.e. $\int P_G(d\psi) = (1/\det(G^{-1}))\det(G^{-1}) = 1$.

We will prove the following statement later.

Lemma 2.5. *With the same notation used so far,*

$$\int P_G(d\psi) \psi_i \bar{\psi}_j = G_{ij}$$

The following are useful facts about Grassmann integrals that we give without proof, we just point out that proving these only involves taking good care of the combinatorics that come from the definition of Grassmann integrals.

Lemma 2.6. *Let $D \in \text{GL}(2, \mathbb{C})$, φ and η two vectors of Grassmann variables. Then, under the following change of variables*

$$\begin{pmatrix} \psi \\ \bar{\psi} \end{pmatrix} = D \begin{pmatrix} \varphi \\ \bar{\varphi} \end{pmatrix} + \begin{pmatrix} \eta \\ \bar{\eta} \end{pmatrix}$$

the integral of a function of Grassmann variables f changes as follows

$$\int \left(\prod_i d\bar{\psi}_i \psi_i \right) f(\psi, \bar{\psi}) = \det(G^{-1}) \int \left(\prod_i d\bar{\varphi}_i \varphi_i \right) f(D(\varphi, \bar{\varphi}) + (\eta, \bar{\eta}))$$

Lemma 2.7. *Let $G \in \text{GL}(N, \mathbb{C})$, then*

$$\int P_G(d\psi) e^{\bar{\eta} \cdot \psi + \bar{\psi} \cdot \eta} = e^{\bar{\eta} \cdot G \eta}$$

Before defining expectations, we need one more operator on a Grassmann algebra to mimic the behaviour of these same integrals done in the complex variables case: derivatives of Grassmann functions. If we are given a polynomial of degree N in N variables, then deriving this polynomial in all of the variables should return the maximal degree term. We will take this as the defining property of derivation, as we defined functions to be nothing more than Taylor series. Now notice that integrating a Grassmann function in all variables (i.e., in all the basis elements) returns exactly the same thing, the coefficient of the maximal degree term.

Definition 2.8 (Grassmann derivatives). Let $f(\psi, \bar{\psi})$ be a Grassmann function, the *Grassmann derivatives* of f with respect to the basis elements are the linear operators defined as

$$\frac{\partial}{\partial \psi_i} f := \int d\psi_i f(\psi, \bar{\psi})$$

Proof (Lemma 2.2). Recalling the “anticommutativity property” of Grassmann integrals from Equation (11), it is clear that the same property holds for derivatives too. We can make use of Grassmann

derivatives to prove this lemma in a way that is closer to the complex variables case. Given a vector of Grassmann variables $\boldsymbol{\eta}$, we can write something similar to (9):

$$\psi_i \bar{\psi}_j e^{-\bar{\psi} \cdot G^{-1} \psi} = - \frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} e^{-\bar{\psi} \cdot G^{-1} \psi + \bar{\eta} \cdot \psi + \bar{\psi} \cdot \eta} \Big|_{\boldsymbol{\eta} = \bar{\boldsymbol{\eta}} = 0}.$$

We can substitute this in the integral to get that

$$\begin{aligned} \int P_G(d\psi) \psi_i \bar{\psi}_j &= - \frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} \int P_G(d\psi) e^{\bar{\eta} \cdot \psi + \bar{\psi} \cdot \eta} \Big|_0 \\ &= - \frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} e^{\bar{\eta} \cdot G \boldsymbol{\eta}} \Big|_0 \\ &= - \frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} (\bar{\eta}_i G_{ij} \eta_j) \Big|_0 \\ &= G_{ij} \end{aligned}$$

which is exactly the thesis. $\square$

We are finally ready to prove that Gaussian Grassmann integrals satisfy Wick's theorem.

Theorem 2.8 (Wick's theorem). *Let $M \in \mathbb{N}$ be the degree of a monomial in Grassmann variables, and let $A = \{a_1, \dots, a_M\}$ and $B = \{b_1, \dots, b_M\}$ be two sets of M indices. The integral of a monomial is*

$$\int P_G(d\psi) \left(\prod_{\alpha \in A, \beta \in B} \psi_\alpha \bar{\psi}_\beta \right) = \sum_{\sigma \in S_N} (-1)^{\text{sgn}(\sigma)} \prod_{j=1}^M G_{a_j, b_{\sigma(j)}}$$

Proof. The proof is almost identical to the one for Theorem (2.3), with the only difference being that now we have to be careful with the sign of the permutations. In particular, deriving the M -th term of the exponential carries an additional factor of $(-1)^M$:

$$\text{l.h.s.} = (-1)^M \left(\prod_{\alpha \in A, \beta \in B} \frac{\partial}{\partial \bar{J}_\alpha} \frac{\partial}{\partial J_\beta} \right) e^{\bar{J} \cdot G J} \Big|_{\mathbf{J} = \bar{\mathbf{J}} = 0}.$$

The rest of the proof follows exactly as in Theorem (2.3). $\square$

At this point, we can take this notion of Gaussian integrals to write the fermionic version of Section 2.1.

2.3 Perturbative series and Feynman diagrams

In this section we are finally going to look at how the order-by-order corrections introduced by a term like $e^{-\beta \lambda V}$ in Equation (8) can, in fact, be written in terms of Feynman diagrams. To do so, we introduce the concept of a truncated expectation. This is the object that ultimately will “contain” the diagrams. Both bosonic and fermionic cases follow the same procedure but present some slight differences here and there. We will state everything in the fermionic case for convenience, and only point out what, if anything, changes in the bosonic case.

Suppose we have an interaction potential V , formally written as a homogeneous polynomial of complex variables. We can represent such a potential as a *vertex*, i.e. a node in a graph, with as many labelled outgoing lines as there are conjugate variables in the expression of V , and as many labelled incoming lines as there are plain variables. For example, our quartic correction of Section 2.1 would

be the formal way of writing a two-body interaction potential $V = \sum_{p,q,k} \Omega_{p,q,k} \bar{\psi}_{q+k} \bar{\psi}_{p-k} \psi_p \psi_q$ and its vertex would be

$$V = \begin{array}{c} \begin{array}{ccc} & p-k & \\ & \nearrow & \\ & \text{---} & \\ & \searrow & \\ & q & \end{array} & \begin{array}{ccc} & q+k & \\ & \nearrow & \\ & \text{---} & \\ & \searrow & \\ & p & \end{array} \end{array} .$$

The *value* of a vertex is defined exactly as the expression of the interaction potential. In a diagram, vertices can be connected by lines called *propagators* or even *contractions*, and as the name suggests the value of such propagators is given by the 2-point function. In particular, if a line labelled by i exits a vertex and enters another as a line labelled by j , then to such a line we associate the value G_{ij} . A vertex may have an outgoing leg connecting back to an incoming line in the same vertex. The *value of a diagram* is obtained by multiplying the values of vertices and propagators following the labelling and contractions dictated by the diagram.

The definitions of the vertices and the propagators involved in the diagrams are called *Feynman rules*. Finally, even though we will not make use of such a definition in our work, a Feynman rule may contain many different “types” of incoming/outgoing lines and associated propagators. This means that in the vertex there might be both fermionic and bosonic variables, and some of them may belong to different sets of variables (for example different Grassmann algebras to represent different families of fermions). Each type of line gets its own propagator, and they should be distinguished graphically.

Example 2. put one of the two first order terms from chapter 3 to explain how to evaluate it.

Definition 2.9 (Truncated expectation). Let $P_G(dz)$

3 Explicit Calculation at Second Order

3.1 Duhamel’s formula

Motivate the study of this system and the scalings $\hbar = N^{-1/3}$ and $\lambda = 1/N$.

The state space for a system of N identical fermions in a box of side length L with periodic boundary conditions is the space of completely antisymmetric L^2 -functions on the torus $\mathbb{T}^{3N} = (\mathbb{R}^3/(2\pi\mathbb{Z}^3))^N$:

$$L_a^2(\mathbb{T}^{3N}) := \{\psi \in L^2(\mathbb{T}^{3N}) : \psi(x_{\sigma(1)}, \dots, x_{\sigma(N)}) = \text{sgn}(\sigma) \psi(x_1, \dots, x_N) \forall \sigma \in S_N\}$$

The Hamiltonian for weakly interacting particles can be written as a sum of the kinetic energy terms plus a sum of two-body interaction terms:

$$H = \hbar^2 \sum_{i=1}^N (-\Delta_{x_i}) + \lambda \sum_{1 \leq i < j \leq N} V(x_i - x_j)$$

where $V : \mathbb{R}^3 \rightarrow \mathbb{R}$ is of class $\mathcal{C}^m$ for some $m \geq 1$ and with period L in all variables, and we assume moreover that $V(x_i - x_j) = V(|x_i - x_j|)$, which implies that $\hat{V}(-k) = \hat{V}(k)$. Working in the grand canonical picture requires lifting this operator to the fermionic Fock space, and requiring that the ground state energy is attained for a fixed number of particles that is greater than zero means that the Hamiltonian picks up an additional term $-\mu\mathcal{N}$, where μ is the *chemical potential*. We absorb the chemical potential in the new definition of the free Hamiltonian as $H_0 = \sum_k (\hbar^2 |k|^2 - \mu) a_k^* a_k$. Putting this all together, the Hamiltonian for this system is written as

$$H = H_0 + \lambda V = \sum_{k \in \mathbb{Z}^3} (\hbar^2 |k|^2 - \mu) a_k^* a_k + \sum_{k,p,q \in \mathbb{Z}^3} \lambda \hat{V}(k) a_{p-k}^* a_{q+k}^* a_q a_p$$

where $a_k^\#$ are creation/annihilation operators on the Fock space that create/annihilate states of definite momentum $k \in \mathbb{Z}^3$.

In the limit of zero temperature for a system of identical fermions, the chemical potential is also the value of the highest energy level occupied. We can therefore express it in terms of the number of particles in the ground state, as this parameter serves as a constraint for such a system through the relation

$$N = \sum_{k \in \mathbb{Z}^3} n_F(\epsilon_k) \xrightarrow{\beta \rightarrow \infty} \sum_{k \in \mathbb{Z}^3} \chi(\epsilon_k \leq \mu) = |\mathbb{B}_{k_F}(0) \cap \mathbb{Z}^3|$$

where k_F is the Fermi momentum, defined by $\epsilon_{k_F} = \mu$. We rederive the Fermi-Dirac distribution in Section (3.2), and justify the zero temperature limit of n_F . This constraint is then just a counting problem for which we can use a well-known argument by Gauss, which gives a lower and an upper bound to the number of points on a lattice inside a sphere. Namely, they are the number of unit cubes that are completely and at least partially inside the sphere, respectively. The result of such bounds can be summed up into the estimate given by the volume of the sphere plus an error proportional to the surface:

$$N = |\mathbb{B}_{k_F} \cap \mathbb{Z}^3| = \frac{4\pi}{3} k_F^3 + \mathcal{O}(k_F^2)$$

which implies that

$$k_F = \left(\frac{3}{4\pi} N \right)^{\frac{1}{3}} + \mathcal{O}(N^0). \quad (12)$$

Since we want to compute the second order contribution to the ground state energy, we will calculate the free energy of the system and take its limit as the temperature goes to zero. Recall that at thermal equilibrium the expectation value of a bounded operator is given by the Gibbs state functional

$$\langle \cdot \rangle_\beta = \frac{\text{Tr}(\cdot e^{-\beta H})}{\text{Tr}(e^{-\beta H})}.$$

The free energy is given by the logarithm of the partition function divided by $-\beta$, and we will compute this logarithm perturbatively. We start such a perturbative calculation up to second order in the operator formalism, but expectation values and higher orders will be dealt with using formal perturbation theory and Grassmann integrals.

Proposition 3.1. *Given $H = H_0 + \lambda V$ as defined above, the following approximation for the free energy holds formally at second order λ , up to an overall factor of $-1/\beta$:*

$$\log \text{Tr} e^{-\beta H} = \log \text{Tr} e^{-\beta H_0} - \beta \lambda \langle V \rangle + \lambda^2 \int_0^\beta d\tau \int_0^\tau d\tau' (\langle V(\tau') V(\tau) \rangle - \langle V(\tau') \rangle \langle V(\tau) \rangle) \quad (13)$$

where Gibbs expectations are understood to be with respect to H_0 at temperature β .

Proof. Start by rewriting the operator $e^{-\beta H} e^{\beta H_0}$ using the fundamental theorem of calculus:

$$\begin{aligned} e^{-\beta H} e^{\beta H_0} &= \mathbb{1} + \int_0^\beta d\tau \left(\frac{d}{d\tau} e^{-\tau H} e^{\tau H_0} \right)(\tau) \\ &= \mathbb{1} + \int_0^\beta d\tau e^{-\tau H} (-H + H_0) e^{\tau H_0} \\ &= \mathbb{1} + \int_0^\beta d\tau e^{-\tau H} e^{\tau H_0} e^{-\tau H_0} (-\lambda V) e^{\tau H_0} \\ &= \mathbb{1} - \lambda \int_0^\beta d\tau e^{-\tau H} e^{\tau H_0} V(\tau) = \star \end{aligned}$$

Note that the integral is well defined as $\tau \mapsto \|e^{-\tau H} e^{\tau H_0} V(\tau)\|$ is a continuous function on the compact domain $[0, \beta]$ and is therefore Lebesgue integrable, which implies that $\tau \mapsto e^{-\tau H} e^{\tau H_0} V(\tau)$ is Bochner integrable. We briefly discuss integration of functions $\mathbb{R}^n \rightarrow \mathfrak{B}(\mathcal{H})$ in the appendix. Now we can see that in the integral we have the same term that appears in the l.h.s. but at a generic temperature τ . Iterating this procedure just once gives us

$$\begin{aligned} \star &= \mathbb{1} - \lambda \int_0^\beta d\tau \left(\mathbb{1} + \lambda \int_0^\tau d\tau' e^{-\tau' H} e^{\tau' H_0} V(\tau') \right) V(\tau) \\ &= \mathbb{1} - \lambda \int_0^\beta d\tau V(\tau) + \lambda^2 \int_0^\beta d\tau \int_0^\tau d\tau' V(\tau') V(\tau) + R(\lambda, V, \beta). \end{aligned}$$

where $R(\lambda, V, \beta)$ is the remainder of this expression. Giving an estimate for the remainder that also allows to prove the convergence of this iterative process is a whole problem of its own, so we will simply truncate the expression and drop the remainder. Now we can invert this relation to isolate $e^{-\beta H}$ and take the trace. Since the trace is a continuous operator from trace class operators to $\mathbb{R}$ and $V e^{-\beta H_0}$ is trace class, we can move it inside the integral (see Appendix A.1 and Appendix A.2) to get:

$$\begin{aligned} \text{Tr } e^{-\beta H} &= \text{Tr } e^{-\beta H_0} - \lambda \int_0^\beta d\tau \text{Tr}(V(\tau) e^{-\beta H_0}) + \lambda^2 \int_0^\beta d\tau \int_0^\tau d\tau' \text{Tr}(V(\tau') V(\tau) e^{-\beta H_0}) \\ &= (\text{Tr } e^{-\beta H_0}) \left(1 - \lambda \int_0^\beta d\tau \langle V(\tau) \rangle + \lambda^2 \int_0^\beta d\tau \int_0^\tau d\tau' \langle (V(\tau') V(\tau)) \rangle \right). \end{aligned}$$

We can now take the logarithm on both sides. Recall the Taylor expansion $\log(1+x) \sim x - x^2/2$ as $x \rightarrow 0$ and apply it to the r.h.s., but note that the first integral gives two contributions (one at first order and one at second order) while the second integral only contributes to the second order. In the end, we are left with:

$$\begin{aligned} \log \text{Tr } e^{-\beta H} &= \log \text{Tr } e^{-\beta H_0} - \lambda \int_0^\beta d\tau \langle V(\tau) \rangle - \frac{1}{2} \left(\lambda \int_0^\beta d\tau \langle V(\tau) \rangle \right)^2 \\ &\quad + \lambda^2 \int_0^\beta d\tau \int_0^\tau d\tau' \langle (V(\tau') V(\tau)) \rangle \\ &= \log \text{Tr } e^{-\beta H_0} - \beta \lambda \langle V \rangle \\ &\quad + \lambda^2 \int_0^\beta d\tau \int_0^\tau d\tau' \left(\langle V(\tau') V(\tau) \rangle - \langle V(\tau') \rangle \langle V(\tau) \rangle \right) \end{aligned} \tag{14}$$

where in the last step we performed the first integral and rewrote the square as a double integral, then taking into account the symmetry under $\tau \longleftrightarrow \tau'$ we could remove the factor $1/2$ by integrating only over the lower triangle $\{(\tau, \tau') \in [0, \beta]^2 : \tau' \leq \tau\}$. $\square$

Remark 23. It is easy to see that taking this iterative process further leads exactly to the definition of the τ -ordered exponential with truncated expectations at all orders, which ultimately shows that, in fact, the perturbative series in the operator formalism and in the Grassmann formalism are formally the same.

3.2 Evaluating expectation values

To introduce Grassmann variables without additional hiccups we need to have a finite number of physical modes in our system. This is achieved by means of a momentum cutoff Λ :

$$\mathcal{D}_\Lambda := \{k \in \mathbb{Z}^3 : |k| < \Lambda\} = \mathbb{Z}^3 \cap \mathbb{B}_\Lambda(0), \quad \text{with } |\mathcal{D}_\Lambda| = M.$$

Let then $\psi = (\psi_1, \dots, \psi_M)$ and $\bar{\psi} = (\bar{\psi}_1, \dots, \bar{\psi}_M)$ be the Grassmann variables associated to the families of operators a and a^* respectively. Having introduced the cutoff, the kinetic part of the Hamiltonian can now be written as a simple finite linear combination of fermionic number density operators, which makes it a bounded operator:

$$H_0 = \sum_{k \in \mathcal{D}_\Lambda} (\epsilon_k - \mu) a_k^* a_k$$

The full physical theory will then be recovered in the limit $\Lambda \rightarrow \infty$.

We begin the calculation by building the gaussian measure for the expectations $\langle \cdot \rangle_{H_0}$. Note that expectations of the form $\langle V(\tau) V(\tau') \rangle_{H_0}$ contains field operators at different temperatures.

Lemma 3.2. *The two-point function for this system has the following form:*

$$G_{k,k'}(\tau, \tau') = \langle a_k^*(\tau) a_{k'}(\tau') \rangle_{\beta, H_0} = \delta_{k,k'} n_F(\epsilon_k) e^{-(\tau - \tau')(\epsilon_k - \mu)} \quad (15)$$

where $n_F(\epsilon) = (e^{\beta(\epsilon - \mu)} + 1)^{-1}$ is the Fermi-Dirac distribution.

Proof. Let us first focus on finding an expression for field operators at a given temperature $a^\#(\tau)$. Recall that in the Heisenberg picture with imaginary time one has that for an operator A its evolution at temperature τ is given by $A(\tau) = e^{-\tau H_0} A e^{\tau H_0}$. Once again we will make use of the fundamental theorem of calculus to write

$$\begin{aligned} e^{-\tau H_0} a_k e^{\tau H_0} &= a_k + \int_0^\tau d\lambda \frac{d}{d\lambda} (e^{-\lambda H_0} a_k e^{\lambda H_0}) \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} (-H_0 a_k + a_k H_0) e^{\lambda H_0} \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} [a_k, H_0] e^{\lambda H_0} \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} \sum_{k' \in \mathcal{D}_\Lambda} (\epsilon_{k'} - \mu) [a_k, a_{k'}^* a_{k'}] e^{\lambda H_0} = \star. \end{aligned}$$

Recall that, for operators A , B and C defined on a common domain, it holds that

$$[A, BC] = \{A, B\}C - B\{A, C\}$$

as can be shown by explicitly writing the expressions for the anticommutators. Using this identity we rewrite the commutator that appears in the last expression in terms of the canonical anticommutator as follows:

$$\begin{aligned} \star &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} \sum_{k' \in \mathcal{D}_\Lambda} (\epsilon_{k'} - \mu) (\{a_k, a_{k'}^*\} a_{k'} - a_{k'}^* \{a_k, a_{k'}\}) e^{\lambda H_0} \\ &= a_k + \int_0^\tau d\lambda e^{-\lambda H_0} (\epsilon_k - \mu) a_k e^{\lambda H_0} \\ &= a_k + (\epsilon_k - \mu) \int_0^\tau d\lambda a_k(\lambda) \end{aligned}$$

where we used that $\{a_k, a_{k'}^*\} = \delta_{k,k'}$ and that $\{a_k, a_{k'}\} = 0$. We now have an expression for $a_k(\tau)$ in terms of its integral, and if we take the derivative with respect to τ we get the following Cauchy problem:

$$\begin{cases} \frac{d}{d\tau} a_k(\tau) = (\epsilon_k - \mu) a_k(\tau) \\ a_k(0) = a_k \end{cases}.$$

By the theorem of existence and uniqueness of the solution to an ODE on a closed interval, it suffices to find one such solution explicitly. This is a well known differential equation:

$$a_k(\tau) = e^{\tau(\epsilon_k - \mu)} a_k(0) = e^{\tau(\epsilon_k - \mu)} a_k.$$

The procedure is completely analogous in the case of $a_k^*(\tau)$ and the result differs only by a sign in the exponent:

$$a_k^*(\tau) = e^{-\tau(\epsilon_k - \mu)} a_k^*(0) = e^{-\tau(\epsilon_k - \mu)} a_k^*.$$

By linearity, we can take the scalar exponential factors out of the expectation value to write the two-point function in terms of the quantity $\langle a_k^* a_{k'} \rangle$:

$$\langle a_k^*(\tau) a_{k'}(\tau') \rangle_{H_0} = e^{-\tau(\epsilon_k - \mu)} e^{\tau'(\epsilon_{k'} - \mu)} \langle a_k^* a_{k'} \rangle_{H_0}.$$

It remains to determine the expectation $\langle a_k^* a_{k'} \rangle$. Recall that the spectrum of the number density operators $a_k^* a_k$ is $\mathbb{N}_0$, and in the fermionic case the occupation number basis decomposes the Fock space like we discussed in Proposition (1.8). Using the cutoff on high momentum, the trace in the Gibbs state functional becomes a sum over a finite amount of states. Let $\{|\mathbf{n}\rangle = |n_1\rangle \otimes \cdots \otimes |n_M\rangle : n_i \in \{0, 1\} \forall i = 1, \dots, M\}$ be the complete orthonormal set of eigenvectors of M independent number density operators (in our case, $M = |\mathcal{D}_\Lambda|$). Calling $Z = \text{Tr } e^{-\beta H_0}$ and using the momenta $k \in \mathcal{D}_\Lambda$ as indices for the basis states labels n_k , we write

$$\begin{aligned} \langle a_k^* a_{k'} \rangle_{\beta, H_0} &= \frac{\text{Tr}(a_k^* a_{k'} e^{-\beta H_0})}{\text{Tr } e^{-\beta H_0}} = \frac{1}{Z} \sum_{\mathbf{n} \in \{0, 1\}^M} \langle \mathbf{n} | e^{-\beta H_0} a_k^* a_{k'} | \mathbf{n} \rangle \\ &= \delta_{k, k'} \frac{1}{Z} \sum_{\mathbf{n} \in \{0, 1\}^M} \langle \mathbf{n} | a_k^* a_k | \mathbf{n} \rangle \prod_{p \in \mathcal{D}_\Lambda} e^{-\beta(\epsilon_p - \mu) n_p} \\ &= \delta_{k, k'} \frac{1}{Z} \sum_{\mathbf{n} \in \{0, 1\}^M} n_k e^{-\beta(\epsilon_k - \mu) n_k} \prod_{p \neq k} e^{-\beta(\epsilon_p - \mu) n_p} \\ &= \delta_{k, k'} \frac{1}{Z} \left(\sum_{n_k=0}^1 n_k e^{-\beta(\epsilon_k - \mu) n_k} \right) \prod_{p \neq k} \sum_{n_p=0}^1 e^{-\beta(\epsilon_p - \mu) n_p} \\ &= \delta_{k, k'} \frac{1}{Z} e^{-\beta(\epsilon_k - \mu)} \prod_{p \neq k} (1 + e^{-\beta(\epsilon_p - \mu)}) \end{aligned}$$

where we used the fact that H_0 is bounded and diagonal in the occupation number basis $\{|\mathbf{n}\rangle\}$, and that $a_k^* a_k |\mathbf{n}\rangle = n_k |\mathbf{n}\rangle$. Using many of the same steps for the denominator, we find that $Z = \prod_{p \in \mathcal{D}_\Lambda} (1 + e^{-\beta(\epsilon_p - \mu)})$, and thus the expression simplifies to

$$\begin{aligned} \langle a_k^* a_{k'} \rangle_{\beta, H_0} &= \delta_{k, k'} \frac{e^{-\beta(\epsilon_k - \mu)} \prod_{p \neq k} (1 + e^{-\beta(\epsilon_p - \mu)})}{\prod_p (1 + e^{-\beta(\epsilon_p - \mu)})} \\ &= \delta_{k, k'} \frac{e^{-\beta(\epsilon_k - \mu)}}{1 + e^{-\beta(\epsilon_k - \mu)}} \\ &= \delta_{k, k'} \frac{1}{e^{\beta(\epsilon_k - \mu)} + 1}. \end{aligned}$$

This is exactly the Fermi-Dirac distribution, and the thesis follows immediately. $\square$

With this quantity we can go over to the computation via Feynman diagrams. Recall the definition for the expectation of a function of Grassmann variables

$$\mathbb{E}_G(\cdot) = \int P_G(d\psi)(\cdot), \quad \text{where } P_G(d\psi) = \frac{1}{\det G} \left(\prod_{j=1}^N d\psi_j d\bar{\psi}_j \right) e^{-\bar{\psi} G^{-1} \psi}$$

and the graphical representation of the interaction term via a vertex:

$$V = \sum_{k,p,q} \hat{V}(k) \bar{\psi}_{p-k} \bar{\psi}_{q+k} \psi_q \psi_p = \begin{array}{c} \begin{array}{cc} p-k & q+k \\ \nearrow & \nearrow \\ \bullet & \\ \nwarrow & \nwarrow \\ q & p \end{array} \end{array}.$$

We can evaluate the expectations that show up in Duhamel's formula separately by following the diagrammatic rules we discussed in the previous chapter. Let's start with $\langle V \rangle_{H_0}$. Omitting the temperatures when writing two-point functions and calling $n_k = n_F(\epsilon_k)$ the calculation is as follows:

$$\begin{aligned} \langle V \rangle_{\beta, H_0} &= \mathbb{E}_G \left(\sum_{k,p,q} \hat{V}(k) \bar{\psi}_{p-k} \bar{\psi}_{q+k} \psi_q \psi_p \right) = \sum_{\Gamma \in \mathcal{G}_{C,1}} \text{val}(\Gamma) \\ &= \begin{array}{c} \text{Diagram 1: A vertex with two incoming lines (bottom) and two outgoing lines (top) forming a figure-eight shape with two loops on the left side.} \end{array} + \begin{array}{c} \text{Diagram 2: A vertex with two incoming lines (bottom) and two outgoing lines (top) forming a figure-eight shape with two loops on the right side.} \end{array} \\ &= \sum_{k,p,q} \hat{V}(k) G_{p-k,p} G_{q+k,q} - \sum_{k,p,q} \hat{V}(k) G_{p-k,q} G_{q+k,p} \\ &= \sum_{k,p,q} (\hat{V}(k) \delta_{k,0} n_p n_q - \hat{V}(k) \delta_{k,p-q} n_p n_q) \\ &= \sum_{p,q} (\hat{V}(0) - \hat{V}(p-q)) n_p n_q. \end{aligned}$$

Let us now consider the term $\langle V(\tau') V(\tau) \rangle - \langle V(\tau') \rangle \langle V(\tau) \rangle$ in the double integral now. This time we have terms with two vertices, and recalling the linked cluster theorem the expression factorizes as follows:

$$\begin{aligned} \langle V(\tau') V(\tau) \rangle - \langle V(\tau') \rangle \langle V(\tau) \rangle &= \mathbb{E}_G(V(\tau') V(\tau)) - \mathbb{E}_G(V(\tau')) \mathbb{E}_G(V(\tau)) \\ &= \mathbb{E}_G^T(V(\tau'); V(\tau)) \\ &= \sum_{\Gamma \in \mathcal{G}_{C,2}} \text{val}(\Gamma) = \star \end{aligned}$$

Since V is such that $\hat{V}(-k) = \hat{V}(k)$, the connected diagrams with two vertices are exactly divided in two groups, up to a renaming of the indices in the resulting sums. Indeed, on one hand we have diagrams that have no loops, i.e. each vertex only has contractions with the other vertex and never with itself, and on the other we have diagrams with two loops. Consider the first group. Fixing an incoming and an outgoing leg on one vertex, there are 4 ways to contract them with legs of the other vertex, and in each case there is only one way to complete the diagram without introducing loops, therefore this group contains 4 diagrams. For the second group, there are 4 ways to form a loop on a vertex, and given a (connected) diagram with two vertices and two loops there exists only one way to complete it. Therefore the second group contains 16 diagrams (all possible ways to put one loop on each vertex). Moreover, all diagrams within their respective groups have the same sign,

since they differ by an even permutation of Grassmann variables.

$$\begin{aligned}
\star &= 4 \tau \text{ (diagram with two vertices } \tau \text{ and } \tau' \text{ connected by four arcs)} + 16 \text{ (diagram with two vertices } \tau \text{ and } \tau' \text{ connected by two arcs)} \\
&= \sum_{\substack{k,p,q \\ k',p',q'}} \hat{V}(k') \hat{V}(k) (4 G_{p-k,p'} G_{q'+k',q} G_{q+k,q'} G_{p'-k',p} + 16 G_{p-k,q} G_{q+k,q'} G_{p'-k',p} G_{q'+k',p'}) \\
&= \sum_{p,q,p'} \left(4 \hat{V}(p-k)^2 e^{(\tau'-\tau)(\epsilon_p+\epsilon_q+\epsilon_k+\epsilon_{p+q-k}-4\mu)} n_p n_q n_k n_{p+q-k} \right. \\
&\quad \left. + 16 \hat{V}(q-k) \hat{V}(p-q) e^{(\tau'-\tau)(2\epsilon_p+\epsilon_q+\epsilon_{q-k}-4\mu)} n_q^2 n_p n_{q-k} \right)
\end{aligned}$$

Renaming some of the indices at both orders and integrating this last expression, as per equation (14), leads us to the following expression for the logarithm:

$$\begin{aligned}
\log \text{Tr} e^{-\beta H} &= \log \text{Tr} e^{-\beta H_0} \\
&\quad - \beta \lambda \sum_{p,k \in \mathcal{D}_\Lambda} \left(\hat{V}(0) - \hat{V}(k) \right) n_p n_{p-k} \\
&\quad - \lambda^2 \sum_{p,q,k \in \mathcal{D}_\Lambda} \left(\frac{4 \hat{V}(k)^2 n_p n_q n_{q+k} n_{p-k}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu} \left(\beta + \frac{1 - e^{-\beta(\epsilon_p+\epsilon_q+\epsilon_{q+k}+\epsilon_{p-k}-4\mu)}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu} \right) \right. \\
&\quad \left. + \frac{16 \hat{V}(q-k) \hat{V}(p-q) n_q^2 n_p n_{q-k}}{2\epsilon_q + \epsilon_p + \epsilon_{q-k} - 4\mu} \left(\beta + \frac{1 - e^{-\beta(2\epsilon_q+\epsilon_p+\epsilon_{q-k}-4\mu)}}{2\epsilon_q + \epsilon_p + \epsilon_{q-k} - 4\mu} \right) \right). \tag{16}
\end{aligned}$$

Now that the computation is complete, we need to prove that removing the UV cutoff, i.e. taking the limit $\Lambda \rightarrow \infty$, leads to a well-defined expression.

So far we have been considering the kinetic energy to be $\epsilon_k = \hbar^2 |k|^2$, but this expression is relevant specifically for the limit $\beta \rightarrow \infty$, while in the following theorem we will consider a more general form with one condition on the asymptotic behaviour at high momentum. The reason for this choice is that, at this level, we are not yet interested in the scaling of this expression in the number of particles, and having the possibility of proving the well-defined-ness of the perturbative series for systems that have a different kinetic term is worth the very little trouble this generalization costs. A quick comment on a slightly less standard notation we will use to state some of the following results. We say that two real or complex sequences $\{x_n\}_n$ and $\{y_n\}_n$ are asymptotic if, and only if, $x_n/y_n \rightarrow 1$. We will indicate that two quantities are asymptotic only up to a proportionality constant by adopting the following notation:

$$x_n \asymp y_n \iff \exists C \neq 0 : \frac{x_n}{y_n} \rightarrow C.$$

To control terms that contain the Fourier transform of the interaction potential we will make use of the following property:

Lemma 3.3 (Non-stationary phase lemma). *If $m \in \mathbb{N}$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $\partial^\alpha f \in L^1(\mathbb{R}^n)$, for all multi-indices α of degree m , then for all such α*

$$|\hat{f}(k)| \leq \frac{1}{|k|^m} \int_{\mathbb{R}^n} dx |\partial^\alpha f(x)|$$

Theorem 3.4. Assume that the interaction potential is such that $\partial^\alpha V \in L^1(\mathbb{T}^{3N})$ for all multi-indices α such that $|\alpha| = m$ for some $m \geq 1$. Assume that ϵ_k depends only on $|k|$ and that it satisfies $\lim_{|k| \rightarrow \infty} \epsilon_k/|k| > 0$. Then the expression in Equation (16) converges in the limit $\Lambda \rightarrow \infty$.

Proof. The zero-th order is trivial, as it doesn't depend on Λ and is, in fact, finite: $e^{-\beta H_0}$ is trace-class and strictly positive definite, which implies $|\log \text{Tr } e^{-\beta H_0}| < \infty$. Before moving on let's discuss the estimates we will use. By the non-stationary phase argument we can conclude that $|\hat{V}(k)| \leq C_1/|k|^m$ for some $C_1 > 0$, thus we have an inverse power decay as $\Lambda \rightarrow \infty$. As for the energy terms, we are assuming that $\epsilon_k \geq C_2|k|$ for all k outside of the ball of radius K for some $K, C_2 > 0$. Finally, the Fermi-Dirac distribution is controlled by an exponential decay in ϵ_k , which is bounded at least by an exponential decay in $|k|$.

Looking at the first order, we can write

$$\begin{aligned} \sum_{p,k \in \mathcal{D}_\Lambda} \left| \left(\hat{V}(0) - \hat{V}(k) \right) n_p n_{p-k} \right| &\leq |\hat{V}(0)| \sum_{p,k \in \mathcal{D}_\Lambda} n_p n_{p-k} + \sum_{p,k \in \mathcal{D}_\Lambda} |\hat{V}(k)| n_p n_{p-k} \\ &= |\hat{V}(0)| \sum_{p \in \mathcal{D}_\Lambda} n_p \sum_{k \in \mathcal{D}_\Lambda} n_{p-k} + \sum_{p \in \mathcal{D}_\Lambda} n_p \sum_{k \in \mathcal{D}_\Lambda} |\hat{V}(k)| n_{p-k}. \end{aligned}$$

By the positivity of the general terms for all $k \in \mathbb{Z}^3$ we can bound the sums by the respective series over $\mathbb{Z}^3$, and because the general terms only depend on the modulus of the momentum we can employ the following strategy. Define the spherical shell $A_n = \{k \in \mathbb{Z}^3 : n \leq |k| < n+1\}$ and write

$$\sum_{k \in \mathbb{Z}^3} a_{|k|} = \sum_{n=0}^{\infty} \sum_{k \in A_n} a_{|k|}.$$

Because $c_1|A_n|a_{n+1} \leq \sum_{k \in A_n} a_{|k|} \leq c_2|A_{n+1}|a_n$ for all n if a_n is decreasing, and because $|A_n| \asymp n^2$, we can conclude that the behaviour of $\sum_{k \in \mathbb{Z}^3} a_{|k|}$ and $\sum_{n=0}^{\infty} n^2 a_n$ is the same. Note that $\sum_{k \in \mathbb{Z}^3} n_k$ converges if, and only if, the series $\sum_{\ell \in \mathbb{N}} \ell^2 e^{-\beta \epsilon_\ell}$ converges, by the asymptotic comparison test. The latter does converge absolutely as shown by the root test:

$$\left| \ell^2 e^{-\beta \epsilon_\ell} \right|^{1/\ell} = e^{\frac{2 \log(\ell) - \beta \epsilon_\ell}{\ell}} \rightarrow Q \in [0, 1).$$

Similarly, $\sum_{k \in \mathbb{Z}^3} |\hat{V}(k)| n_k$ converges, as

$$\sum_{k \in \mathbb{Z}^3} |\hat{V}(k)| n_k \leq \sum_{k \in \mathbb{Z}^3} \frac{C_1 n_k}{|k|^m} \leq \sum_{k \in \mathbb{Z}^3} C_1 n_k.$$

Thus, in the limit in $\Lambda \rightarrow \infty$, the first order is an absolutely convergent series.

Very similar arguments can be used to show that the second order has an absolutely convergent series as a limit. Indeed, we can bound the first term in the sum (the second order direct term) by

$$\sum_{p,q,k \in \mathcal{D}_\Lambda} \hat{V}(k)^2 (\dots) \leq \sum_p \frac{4n_p}{\epsilon_p - 4\mu} \sum_q n_q \sum_k \hat{V}(k)^2 n_{q+k} n_{p-k} \left(\beta + \frac{1}{\epsilon_p - 4\mu} \right),$$

and turning once more sums over $\mathbb{Z}^3$ into series over spherical shells, this is absolutely convergent, because the additional factor $1/\epsilon_p$ does not change the result of any of the comparison tests done before. The exact same argument can be applied to the other term (second order exchange term), and this concludes the proof. $\square$

We can then essentially replace the sums over $\mathcal{D}_\Lambda$ in equation (16) with series over $\mathbb{Z}^3$. Inserting

the factor of $-1/\beta$ to get the free energy leads us to its complete expression:

$$\begin{aligned}
F &= F^{(0)} + F_D^{(1)} + F_X^{(1)} + F_D^{(2)} + F_X^{(2)} \\
&= F^{(0)} + \lambda \sum_{p,k \in \mathbb{Z}^3} \hat{V}(0) n_p n_{p-k} - \lambda \sum_{p,k \in \mathbb{Z}^3} \hat{V}(k) n_p n_{p-k} \\
&\quad + \lambda^2 \sum_{p,q,k \in \mathbb{Z}^3} \frac{4\hat{V}(k)^2 n_p n_q n_{q+k} n_{p-k}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu} \left(1 + \beta^{-1} \frac{1 - e^{-\beta(\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu)}}{\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu} \right) \\
&\quad + \lambda^2 \sum_{p,q,k \in \mathbb{Z}^3} \frac{16\hat{V}(k-q)\hat{V}(p-q)n_q^2 n_p n_{q-k}}{2\epsilon_q + \epsilon_p + \epsilon_{q-k} - 4\mu} \left(1 + \beta^{-1} \frac{1 - e^{-\beta(2\epsilon_q + \epsilon_p + \epsilon_{q-k} - 4\mu)}}{2\epsilon_q + \epsilon_p + \epsilon_{q-k} - 4\mu} \right) \quad (17)
\end{aligned}$$

where each term has been named appropriately following the order in which they appear: $F^{(0)}$ is the free energy of the unperturbed system (no interactions), $F_D^{(i)}$ is the direct term at order i and $F_X^{(i)}$ is the exchange term at order i .

3.3 Mean field scaling

With the convergence of the free energy under control, the next step is to look at the limit $\beta \rightarrow \infty$ to retrieve the ground state energy and complete our perturbative proof of the **RPA scaling in the number of particles N** .

Lemma 3.5. *With the same hypotheses as in Theorem (3.4), all the series in Equation (17) converge uniformly in β over $[\delta, \infty)$ for some $\delta > 0$, and the limit $\beta \rightarrow \infty$ can thus be taken inside the sums.*

Proof. The only troubling factor is the Fermi-Dirac distribution, as it has a discontinuity at $\epsilon_k = \mu$ in the limit $\beta \rightarrow \infty$. By splitting each sum into a finite part up to, say, $\epsilon_k = 2\mu$, and a series over the remaining values of k , we can avoid this discontinuity. Moreover, we are restricting the domain on which to check for uniform convergence to $[\delta, \infty)$ for some $\delta > 0$ because we are considering a limit away from $\beta = 0$. With these things in mind we can see that the series $\sum_k n_k$ converges uniformly, as

$$\sum_{k: \epsilon_k \geq 2\mu} n_k \leq \sum_k \sup_{\beta \in [\delta, \infty)} n_k \leq \sum_k e^{-\delta(\epsilon_k - \mu)}$$

and the upper bound converges absolutely by the same arguments used in the previous theorem. Modifying this series with a prefactor of $1/(\epsilon_k - \mu)$, $\hat{V}(k)$ or $\hat{V}^2(k)$ in the general term does not change the result, as these factors do not depend on β and still lead to an absolutely convergent series as an upper bound. We can see that, by rewriting sums of the form $\sum_{p,q,k}(\dots)$ as $\sum_q(\dots \sum_p(\dots \sum_k(\dots)))$, we only need to check the convergence of the innermost series in all the terms of Equation (17), and the rest will follow from arguments that we already covered. Consider the terms $F_D^{(1)} + F_X^{(1)}$:

$$\begin{aligned}
\sum_{p,k} \left| \hat{V}(0) - \hat{V}(k) \right| n_p n_{p-k} &= \sum_p n_p \sum_k \left| \hat{V}(0) - \hat{V}(k) \right| n_{p-k} \\
&\leq \sum_p n_p \sum_k \left| \hat{V}(0) - \hat{V}(k) \right| \sup_{\beta \in [\delta, \infty)} n_{p-k} = \star
\end{aligned}$$

but the series containing $\sup_{\beta}(\dots)$ converges to some $C > 0$ as we discussed above, and we can therefore write $\star = \sum_p C n_p$, which again converges. For the term $F_D^{(2)}$, we can identify two terms that need our attention, up to trivial bounds:

$$A := \sum_k \frac{\hat{V}(k)^2 n_{q+k} n_{p-k}}{\epsilon_{q+k} + \epsilon_{p-k} - 2\mu} \quad \text{and} \quad B := \sum_k \left| \frac{\hat{V}(k)^2 n_{q+k} n_{p-k} (1 - e^{-\beta(\epsilon_p + \epsilon_q + \epsilon_{q+k} + \epsilon_{p-k} - 4\mu)})}{\beta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)^2} \right|.$$

The exchange term $F_X^{(2)}$ will exhibit the same behaviour, both in terms of dependence on β and in terms of pointwise convergence. The series A is very simple: we bound the Fermi-Dirac distributions with their supremum and notice that

$$A \leq \sum_k \frac{4\hat{V}(k)^2 e^{-\delta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)}}{\epsilon_{q+k} + \epsilon_{p-k} - 2\mu}.$$

Let's now discuss B . Distributing the product at the numerator, the first term is a product of Fermi-Dirac distributions divided by β , which means it is monotonically decreasing in β for all q, p, k outside the Fermi ball. As for the exponential term it is easy to check that the resulting general term is monotonically decreasing in β far enough from the origin, as the only extremal point is a maximum that moves towards the origin as k grows. Therefore, once again, the supremum is found at $\beta = \delta$ and we are left with

$$B \leq \sum_k \frac{\hat{V}(k)^2 e^{-\delta(\epsilon_{q+k} + \epsilon_{p-k} - 2\mu)}}{\delta(\epsilon_{q+k} + \epsilon_{p-k})^2} + \sum_k \frac{\hat{V}(k)^2 e^{-\delta(\epsilon_p + \epsilon_q + 2\epsilon_{q+k} + 2\epsilon_{p-k} - 6\mu)}}{\delta(\epsilon_{q+k} + \epsilon_{p-k})^2},$$

which finally shows that the second order is uniformly convergent too, completing the proof. $\square$

Now that we have proved that we can take the limit of zero temperature inside the series, we are left with the task of computing it. The zero temperature Fermi-Dirac distribution is just the characteristic function of the Fermi ball B_F :

$$n_F(\epsilon) \xrightarrow{\beta \rightarrow \infty} \chi(\epsilon \leq \mu).$$

At last, using this fact we arrive at the ground state energy scaling in N , which is just a couple of estimates away. We will assume a stronger condition on the potential, namely that its Fourier transform has compact support, but we may as well choose to assume that $\hat{V}(k) \sim e^{-|k|}$.

Theorem 3.6. *Assuming $\text{supp}(\hat{V})$ is compact, the ground state energy of a system of weakly interacting fermions exhibits the following scalings in the number of particles N at second order in perturbation theory:*

- $F^{(0)} = \mathcal{O}(N)$,
- $F_D^{(1)} = \mathcal{O}(N)$ and $F_X^{(1)} = \mathcal{O}(N^0)$,
- $F_D^{(2)} = \mathcal{O}(N^{-\frac{1}{3}})$ and $F_X^{(2)} = \mathcal{O}(N^{-\frac{1}{3}})$ *CHECK SPECIFIC POWER OF N for $F_{2,X}$.*

Proof. Of course we proceed order by order, starting from the unperturbed free energy to make sure that everything is in order. Recall that $B_F = \{p \in \mathbb{Z}^3 : |p| < k_F\}$ and expand the expression for $F^{(0)}$ as

$$F_0 = -\frac{\log(Z)}{\beta} = -\sum_{p \in \mathbb{Z}^3} \frac{1}{\beta} \log(1 + e^{-\beta(\epsilon_p - \mu)}) = \sum_{p \in B_F} (\epsilon_p - \mu) + \sum_{p \notin B_F} \log(1 + e^{-\beta(\epsilon_p - \mu)}),$$

where in the limit the second term contributes to order $\mathcal{O}(N^0)$. From here we can change the index summed over in $\sum_{|p| \leq CN^{1/3}}$ to approximate the result with an integral which contributes to order $\mathcal{O}(N^0)$. Recall that $\hbar = N^{-1/3}$ and $k_F = \mathcal{O}(N^{1/3})$, then take $p \mapsto p' = N^{-1/3}p$, and since in the Fermi ball one has $|p| \leq CN^{1/3}$ this leads to $|p'| \leq C$. This allows for the following known

approximation by integration:

$$\begin{aligned}
\sum_{p \in B_F} (\hbar^2 p^2 - \mu) &= N^{-\frac{2}{3}} \sum_{p \in \mathbb{Z}^3: |k| \leq C N^{1/3}} p^2 - \mu \sum_{p \in B_F} 1 \\
&= N^{-\frac{2}{3}} N \left(\frac{1}{N} \sum_{p' \in \mathbb{Z}^3: |p'| \leq C} (N^{\frac{1}{3}} p')^2 \right) - N\mu \\
&= N \left(\int_{|p| \leq C} d^3 p p^2 + \mathcal{O}(N^{-\frac{1}{3}}) \right) - N\mu \\
&= \tilde{C} N + \mathcal{O}(N^{\frac{2}{3}}) = \mathcal{O}(N)
\end{aligned}$$

where we used that, from Equation (12), $\mu = \hbar^2 k_F^2 = \mathcal{O}(N^0)$. This shows that indeed $F^{(0)} \asymp N$.

Onto the first order, recall the scaling limit $\lambda = N^{-1}$ to see that

$$\begin{aligned}
F_D^{(1)} + F_X^{(1)} &= \lambda \sum_{p, k \in \mathbb{Z}^3} (\hat{V}(0) - \hat{V}(k)) n_p n_{p-k} = \lambda \sum_{p, k \in \mathbb{Z}^3} (\hat{V}(0) - \hat{V}(k)) n_p n_{p-k} \\
&\xrightarrow{\beta \rightarrow \infty} N^{-1} \hat{V}(0) \left(\sum_{p, k \in B_F} 1 \right) - N^{-1} \sum_{\substack{p, k \in B_F: \\ p-k \in B_F}} \hat{V}(k) \\
&= \hat{V}(0) N + N^{-1} \mathcal{O}(N) = \hat{V}(0) N + \mathcal{O}(N^0).
\end{aligned}$$

where in the last steps we used the condition that $\text{supp}(\hat{V})$ is compact, so that $\sum_k \hat{V}(k) = \mathcal{O}(N^0)$ and $\sum_{q, k} \hat{V}(k) = \mathcal{O}(N)$. With this expression we can conclude that the first order indeed contributes to order $\mathcal{O}(N)$ with the direct term and to order $\mathcal{O}(N^0)$ with the exchange term.

To tackle the second order we use the same strategies that worked for the zero-th and first orders. Note that, when using the integral approximation for the sums over p and q , the domain of integration will carry an order of $N^{-1/3}$ for each integral, because it involves imposing that some momentum $N^{-1/3} p$ and its translation $N^{-1/3} p - k$ must be inside a sphere of constant radius, where k ranges over the entire Fermi ball. Calling $C = (3/4\pi)^{1/3}$, the calculation for the direct term is done as follows:

$$\begin{aligned}
F_D^{(2)} &\xrightarrow{\beta \rightarrow \infty} \lambda^2 \sum_{k \in \mathbb{Z}^3} \sum_{\substack{p, q \in B_F: \\ q+k, p-k \in B_F}} \frac{\hat{V}(k)^2}{\hbar^2 (p^2 + q^2 + (q+k)^2 + (p-k)^2 - 4k_F^2)} \\
&= N^{-2} \sum_{k \in \mathbb{Z}^3} \sum_{\substack{p, q \in B_F: \\ q+k, p-k \in B_F}} N^{\frac{2}{3}} \frac{\hat{V}(k)^2}{p^2 + q^2 + (q+k)^2 + (p-k)^2 - 4C^2 N^{\frac{2}{3}}} \\
&= N^{\frac{2}{3}} \sum_{k \in \mathbb{Z}^3} \frac{1}{N} \sum_{\substack{q \in N^{-1/3} \mathbb{Z}^3: \\ |q| \leq C \\ |q+N^{-1/3}k| \leq C}} \frac{1}{N} \sum_{\substack{p \in N^{-1/3} \mathbb{Z}^3: \\ |p| \leq C \\ |p-N^{-1/3}k| \leq C}} \frac{\hat{V}(k)^2}{p^2 + q^2 + (q+k)^2 + (p-k)^2 - 4C^2 N^{\frac{2}{3}}} \\
&= \sum_{k \in \mathbb{Z}^3} \iint_{\substack{|q|, |p| \leq C \\ |q+N^{-1/3}k| \leq C \\ |p-N^{-1/3}k| \leq C}} d^3 p d^3 q \frac{\hat{V}(k)^2}{p^2 + q^2 + (q+N^{-\frac{1}{3}}k)^2 + (p-N^{-\frac{1}{3}}k)^2 - 4C^2} + \mathcal{O}(N^{-\frac{1}{3}}).
\end{aligned}$$

Looking at this expression we can see that, as we observed before, each integral contributes to order $\mathcal{O}(N^{-1/3})$ to an overall order $\mathcal{O}(N^{-2/3})$, which is subleading with respect to the integral

approximation error $\mathcal{O}(N^{-1/3})$. The integrand doesn't add to the leading order in N . All in all we are left with $\mathcal{O}(N^{-1/3})$.

Very similar steps can be taken to compute the scaling order of the term $F_X^{(2)}$. The difference is that now only one sum needs to be approximated with an integral, and this is already enough to say that $F_X^{(2)}$ will be of subleading order with respect to $F_D^{(2)}$. Indeed:

$$\begin{aligned}
F_X^{(2)} &\xrightarrow{\beta \rightarrow \infty} \lambda^2 \sum_{k \in \mathbb{Z}^3} \sum_{p \in B_F} \sum_{\substack{q \in B_F: \\ q-k \in B_F}} \frac{\hat{V}(k-q)\hat{V}(p-q)}{\hbar^2(2q^2 + p^2 + (q-k)^2 - 4k_F^2)} \\
&= N^{-1} \sum_{k \in \mathbb{Z}^3} \sum_{p \in B_F} \frac{1}{N} \sum_{\substack{q \in N^{-1/3}\mathbb{Z}^3: \\ |q| \leq C \\ |q-N^{-1/3}k| \leq C}} \frac{\hat{V}(k-q)\hat{V}(p-q)}{N^{-\frac{2}{3}}(2N^{\frac{2}{3}}q^2 + p^2 + (N^{\frac{1}{3}}q+k)^2 - 4C^2N^{\frac{2}{3}})} \\
&= \text{NOPE}.
\end{aligned}$$

□

4 On higher orders of perturbation theory

- comment on the regularity condition $\partial^\alpha \in L^1$ which excludes coulomb interaction;
- comment on third order somewhat explicitly;
- discuss the hope of proving that at every order n the direct term $\hat{V}(k)^n$ scales with $N^{-1/3}$;
- discuss difficulties in proving that the rest in duhamel's formula goes to zero and that non-direct terms converge.

A More about Hilbert spaces

We group in this appendix a series of definitions and theorems that are not necessarily part of the basic foundations of quantum mechanics, but are still crucial in understanding how exactly some computations are carried out and why they hold true. Some of the results presented here go beyond the scope of our work as far as their proof is concerned, and only serve the purpose of painting a clearer picture of what some tools actually are without just referencing an entire course on functional analysis.

A.1 Special operators

Definition A.1 (Self-adjoint operators). An operator on a Hilbert space $A : D(A) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ is called *self-adjoint* if $A = A^*$.

Note that this condition is stronger than being symmetric.

Definition A.2 (Compact operators). An operator on a Hilbert space $A : D(A) \subseteq \mathcal{H} \rightarrow \mathcal{H}$ is called *compact* if $D(A) = \mathcal{H}$ and there exist complete orthonormal systems $\{u_i : i \in \mathbb{N}\}$ and $\{v_i : i \in \mathbb{N}\}$ for $\mathcal{H}$ and a sequence $\{\lambda_i : i \in \mathbb{N}\}$ such that $\lambda_i \rightarrow 0$ such that

$$A\phi = \sum_{n=0}^{\infty} \lambda_n \langle v_n, \phi \rangle u_n$$

for all $\phi \in \mathcal{H}$.

Definition A.3 (Trace class and Hilbert-Schmidt operators). A compact operator A is said to be *trace class* if $\sum_{n=1}^{\infty} |\lambda_n| < \infty$ and is said to be *Hilbert-Schmidt* if $\sum_{n=1}^{\infty} |\lambda_n|^2 < \infty$. For a trace class operator A we define its *trace* as

$$\mathrm{Tr}(A) := \sum_{n=1}^{\infty} \lambda_n \langle v_n, u_n \rangle.$$

Proposition A.1. *The trace of a trace class operator A is finite, and given an orthonormal basis $\{\phi_n\}_{n \in \mathbb{N}}$ for $\mathcal{H}$ the trace is*

$$\mathrm{Tr}(A) = \sum_{n=1}^{\infty} \langle \phi_n, A\phi_n \rangle.$$

Remark 24. The rigorous derivation of the quantities used in this work relies heavily on the properties of the operator $e^{-\beta H}$, given H a self-adjoint operator (in our case H is also bounded, but spectral theory can extend this to the unbounded case). This operator is trace-class, and it is strictly positive definite. Moreover, if V is a bounded operator defined on the image of $e^{-\beta H}$ then it also holds that $Ve^{-\beta H}$ is trace class.

Definition A.4 (Projection). A linear operator P on a Hilbert space $\mathcal{H}$ which is self-adjoint ($P^* = P$) and idempotent ($P^2 = P$) is called a *projection operator* or *projector*.

Proposition A.2. *Given a Hilbert space $\mathcal{H}$, projectors defined on $\mathcal{H}$ have the following properties:*

- a) *every projector P is such that $P \neq 0$ and $\|P\| = 1$;*
- b) *P is a projector if, and only if, $\mathbb{1} - P$ is a projector;*
- c) *if P is a projector, then*

$$\mathcal{H} = \mathrm{ran} P \oplus \ker P, \quad P|_{\mathrm{ran} P} = \mathbb{1}, \quad P|_{\ker P} = 0.$$

In particular, the spectrum of P is $\sigma(P) = \{0, 1\}$;

- d) *there is a bijection between projectors defined on $\mathcal{H}$ and closed subspaces of $\mathcal{H}$.*
- e) *if $\{e_1, \dots, e_N\}$ for $N \in \mathbb{N}$ or $N = \infty$ is an orthonormal system in $\mathcal{H}$, then the projection P_M on $M = \mathrm{span}\{e_1, \dots, e_N\}$ is*

$$P_M x = \sum_{n=1}^N \langle e_n, x \rangle e_n, \quad \forall x \in \mathcal{H}.$$

Definition A.5 (Isometries and Unitaries). Given two Hilbert spaces $\mathcal{H}$ and $\mathcal{K}$, a linear map $U : \mathcal{H} \rightarrow \mathcal{K}$ with the property that

$$\|Ux\|_{\mathcal{K}} = \|x\|_{\mathcal{H}} \quad \forall x \in \mathcal{H}$$

is called an *isometry* between $\mathcal{H}$ and $\mathcal{K}$. A surjective isometry is called a *unitary operator*.

Unitary operators are of particular interest in quantum mechanics, as the property of preserving the inner product means that these are the natural objects to describe operations that need to preserve probability amplitudes.

Proposition A.3. *For a bounded linear operator between Hilbert spaces $U : \mathcal{H} \rightarrow \mathcal{K}$, the following statements are equivalent:*

- a) U is a unitary operator;
- b) $U^*U = \mathbb{1}_{\mathcal{K}}$ and $UU^* = \mathbb{1}_{\mathcal{H}}$;
- c) $U\mathcal{H} = \mathcal{K}$ and $\langle x, y \rangle_{\mathcal{H}} = \langle Ux, Uy \rangle_{\mathcal{K}}$, for all $x, y \in \mathcal{H}$.

Recall that the dual of a complex Hilbert space $\mathcal{H}$ is the Hilbert space $\mathcal{H}^* = \{\varphi : \mathcal{H} \rightarrow \mathbb{C} : \varphi \text{ is a continuous linear functional}\}$, where continuity is with respect to the norm topology on $\mathcal{H}$. To conclude this part of our discussion on the basics of operators on a Hilbert space, we give three more results whose importance we will not discuss further, as it goes beyond the scope of our work.

Theorem A.4 (Riesz representation theorem). *Let $\mathcal{H}$ be a complex Hilbert space. For every $x \in \mathcal{H}$ the map*

$$\begin{aligned} F : \mathcal{H} &\longrightarrow \mathbb{C} \\ y &\longmapsto F(y) := \langle x, y \rangle \end{aligned}$$

defines a continuous functional, and thus $F \in \mathcal{H}^$. Conversely, for every $F \in \mathcal{H}^*$ there exists a unique $x_F \in \mathcal{H}$ such that $F(y) = \langle x_F, y \rangle$ for all $y \in \mathcal{H}$. Moreover, it holds that $\|x_F\|_{\mathcal{H}} = \|F\|_{\mathcal{H}^*}$.*

Remark 25. The map $J : \mathcal{H} \rightarrow \mathcal{H}^*$ that sends a vector to its uniquely identified continuous functional on $\mathcal{H}$ defines a canonical “isomorphism” between a Hilbert space and its dual, which means Hilbert spaces are self-dual. Technically J is not even a homomorphism since it is conjugate-linear rather than linear, and calling it an isomorphism is a bit imprecise, but this does not pose any problem to us.

Theorem A.5 (Spectral theorem for self-adjoint operators). *Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be a self-adjoint operator on the Hilbert space $\mathcal{H}$ over the field $\mathbb{K}$. Then there exists a measure space (X, Σ, μ) , a unitary operator $U : L^2(X, d\mu) \rightarrow \mathcal{H}$ and a function $f : X \rightarrow \mathbb{K}$ such that*

$$A = UM_fU^*$$

where M_f is the multiplication by f , i.e. the operator defined by $(M_f\psi)(x) = f(x)\psi(x)$.

Remark 26. In the case where A is also compact, then the spectrum $\sigma(A)$ is discrete and the function f could be seen as mapping an index over $\sigma(A)$ to an eigenvalue, and the thesis would read as $(M_f\psi)_i = f_i\psi_i$, which tells us that ψ_i is an eigenvector with eigenvalue f_i . This special case makes it easier to understand this version of the spectral theorem as a generalization to the case of self-adjoint operators with possibly continuous spectrum.

Theorem A.6 (C^* -algebra of bounded operators). *Let $\mathcal{H}$ be a Hilbert space. The set*

$$\mathfrak{B}(\mathcal{H}) = \{A : \mathcal{H} \rightarrow \mathcal{H} : A \text{ is bounded} \}$$

is a Banach space, i.e. a complete normed space, via the usual definition of addition between operators (assuming the domain is the entire space $\mathcal{H}$) and multiplication by a scalar. It is also a C^ -algebra, i.e. a normed algebra with an involution $*$, via the adjoint of bounded operators.*

A.2 Integration of operator-valued functions

We want to define integration for functions of the type $f : X \rightarrow \mathcal{B}$, where (X, Σ, μ) is a measure space and $\mathcal{B}$ is a Banach space (while differentiation is automatic if the measure space is $\mathbb{R}^n$ via the usual definition as limit of the incremental ratio). First we define *simple functions*: let $\{E_1, \dots, E_n\}$ be elements of the σ -algebra Σ and $\{b_1, \dots, b_n\}$ elements of $\mathcal{B}$, then the function

$$s(x) := \sum_{i=1}^n \chi_{E_i} b_i$$

is called a simple function and its integral is defined as

$$\int_X s d\mu := \sum_{i=1}^n \mu(E_i) b_i.$$

Definition A.6 (Bochner integral). A measurable function $f : X \rightarrow \mathcal{B}$ is called *Bochner integrable* if there exists a sequence of simple functions $\{s_n : n \in \mathbb{N}\}$ such that

$$\lim_{n \rightarrow \infty} \int_X \|f - s_n\|_{\mathcal{B}} d\mu = 0,$$

i.e. if its point-wise norm is Lebesgue integrable. In this case, the *integral of f* is defined as

$$\int_X f d\mu := \lim_{n \rightarrow \infty} \int_X s_n d\mu.$$

It can be shown that $\{\int_X s_n d\mu : n \in \mathbb{N}\}$ is a Cauchy sequence, hence the integral is well-defined.

Remark 27. The integration rules for “nice” functions like powers and exponentials are identical to the case of scalar functions, just like for differentiation, but we will not prove this.

Proposition A.7. *Bochner integration satisfies the following properties:*

1. a function $f : X \rightarrow \mathcal{B}$ is Bochner integrable if, and only if, $\int_X \|f\|_{\mathcal{B}} d\mu < \infty$;
2. if $T : \mathcal{B} \rightarrow \mathcal{B}'$ is a continuous linear operator between Banach spaces and $f : X \rightarrow \mathcal{B}$ is Bochner integrable, then Tf is also Bochner integrable and it holds that

$$T \int_E f d\mu = \int_X T f d\mu$$

for all $E \in \Sigma$;

3. (Dominated convergence theorem) if $\{f_n\}$ is a sequence of Bochner integrable functions that converges almost everywhere to a limit function f , and if there exists an integrable function g such that $\|f_n(x)\|_{\mathcal{B}} \leq g(x)$ almost everywhere on X , then it holds that

$$\int_E \|f - f_n\|_{\mathcal{B}} d\mu \rightarrow 0, \quad \text{and} \quad \int_E f_n d\mu \rightarrow \int_E f d\mu$$

for all $E \in \Sigma$;

4. (Fundamental theorem of calculus) if $f : X \rightarrow \mathcal{B}$ is Bochner integrable over $[a, b] \subseteq \mathbb{R}$ and $F : X \rightarrow \mathcal{B}$ is an antiderivative for f , then

$$\int_{[a, b]} f d\mu = F(b) - F(a).$$

Remark 28. In particular, since functionals in the dual of a Hilbert space are also continuous operators between Banach spaces, given $f : X \rightarrow \mathcal{B}(\mathcal{H})$ a Bochner integrable function, it holds that

$$\langle \phi, \left(\int_X f d\mu \right) \psi \rangle_{\mathcal{H}} = \int_X \langle \phi, f \psi \rangle_{\mathcal{H}} d\mu.$$

Remark 29. An important property, which we make use of in our work, is that the set of trace-class operators is a Banach space with the trace norm, and the trace is a continuous operator from this space to the base field. This means that, because of the above proposition, whenever we take the trace of an integral of an operator-valued function, we can take the trace inside the integral.

With this notion of integration (and with the usual notion of differentiation) we can give an important theorem on ordinary differential equations that we used in our work. The proof of this theorem is completely analogous to the case where the Banach space is just $\mathbb{R}^n$, with the only difference being that the point-wise norm of a function is given by the norm on $\mathcal{B}$.

Theorem A.8 (Existence and uniqueness of a solution to an ODE). *Let $f : [a, b] \times \mathcal{B} \rightarrow \mathcal{B}$ be a Lipschitz continuous function on $[a, b]$ uniformly on $\mathcal{B}$, and let it define the following Cauchy problem for $(x_0, y_0) \in [a, b] \times \mathcal{B}$:*

$$\begin{cases} y'(x) = f(x, y(x)) \\ y(x_0) = y_0 \end{cases}$$

where $y : [a, b] \rightarrow \mathcal{B}$. Then there exists a unique solution for the Cauchy problem over $[a, b]$.

A.3 Direct sums and tensor products of Hilbert spaces

Recall that, given two vector spaces V and W over a field $\mathbb{K}$, their *direct sum* is the set $V \times W$ endowed with a vector space structure defined via the following operations:

$$\begin{aligned} (v_1, w_1) + (v_2, w_2) &:= (v_1 + v_2, w_1 + w_2), \\ \lambda(v, w) &:= (\lambda v, \lambda w) \end{aligned}$$

for all $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$ and $\lambda \in \mathbb{K}$, and is denoted $V \oplus W$. Suppose now $\mathcal{H}_1$ and $\mathcal{H}_2$ are Hilbert spaces, then we have the natural result

Proposition A.9. *If $\langle \cdot, \cdot \rangle_i$ denotes the inner product on $\mathcal{H}_i$ for $i = 1, 2$, then the bilinear $\langle \cdot, \cdot \rangle$ given by*

$$\langle (x_1, x_2), (y_1, y_2) \rangle := \langle x_1, y_1 \rangle_1 + \langle x_2, y_2 \rangle_2$$

defines an inner product with respect to which the space $\mathcal{H}_1 \oplus \mathcal{H}_2$ is complete. Thus $(\mathcal{H}_1 \oplus \mathcal{H}_2, \langle \cdot, \cdot \rangle)$ is a Hilbert space.

We can extend this definition of direct sum to the case where we consider a countable set of Hilbert spaces $\{\mathcal{H}_i : i \in \mathbb{N}\}$ over the same field $\mathbb{K}$.

Proposition A.10. *The set $\mathcal{H}$ of sequences $x = \{x_i : x_i \in \mathcal{H}_i, i \in \mathbb{N}\}$ that satisfy*

$$\sum_{i=1}^{\infty} \|x_i\|_i^2 < \infty$$

is a Hilbert space with the operations $\{x_i\} + \{y_i\} := \{x_i + y_i\}$, $\lambda\{x_i\} := \{\lambda x_i\}$ and $\langle \{x_i\}, \{y_i\} \rangle := \sum_{i=1}^{\infty} \langle x_i, y_i \rangle_i$. This Hilbert space is denoted

$$\mathcal{H} = \bigoplus_{i=1}^{\infty} \mathcal{H}_i.$$

To build the tensor product we will follow an algebraic approach which involves a similar procedure but with additional steps. Given two vector spaces V and W over a field $\mathbb{K}$, define the vector space of finite linear combinations of ordered pairs in $V \times W$ as

$$\Lambda(V, W) := \left\{ \sum_{i=1}^N a_i(x_i, y_i) : a_i \in \mathbb{K}, x_i \in V, y_i \in W \forall i = 1, \dots, N \right\}.$$

On this space we impose algebraic relations through the quotient space

$$V \otimes W := \Lambda(V, W) / \Lambda_0$$

where Λ_0 is the subspace generated by elements

$$\begin{aligned} & (x, y_1 + y_2) - (x, y_1) - (x, y_2), \\ & (x_1 + x_2, y) - (x_1, y) - (x_2, y), \\ & (\lambda x, y) - \lambda(x, y), \\ & (x, \lambda y) - \lambda(x, y) \end{aligned}$$

for all $x, x_1, x_2 \in V$ and $y, y_1, y_2 \in W$. If $\chi : \Lambda(V, W) \rightarrow \Lambda(V, W) / \Lambda_0$ is the projection to the quotient space, then we denote the elements in the quotient as $x \otimes y = \chi((x, y))$. Once again we can now extend this definition to the case of Hilbert spaces.

Proposition A.11. *Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces over the same field. If $\langle \cdot, \cdot \rangle_i$ denotes the inner product on $\mathcal{H}_i$ for $i = 1, 2$, then the bilinear $\langle \cdot, \cdot \rangle$ given by*

$$\langle x_1 \otimes x_2, y_1 \otimes y_2 \rangle := \langle x_1, y_1 \rangle_1 \langle x_2, y_2 \rangle_2$$

defines an inner product on $\mathcal{H}_1 \otimes \mathcal{H}_2$. The metric completion $(\mathcal{H}_1 \tilde{\otimes} \mathcal{H}_2, \langle \cdot, \cdot \rangle)$, unique up to isomorphism, is a Hilbert space, called tensor product of Hilbert spaces.

Remark 30. Note that, in general, the metric completion is required, as the algebraic tensor product by itself may not be complete with respect to the norm induced by the inner product defined in the proposition. However, with a slight abuse of notation we still denote $\mathcal{H}_1 \otimes \mathcal{H}_2$ the tensor product of the Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$.

In the case where $\mathcal{H}_1$ and $\mathcal{H}_2$ are separable Hilbert spaces, we can give an explicit construction of their tensor product. Indeed, let $\{u_i : i \in \mathbb{N}_1\}$ and $\{v_i : i \in \mathbb{N}\}$ be orthonormal systems for $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively. Then it holds that

$$\mathcal{H}_1 \otimes \mathcal{H}_2 = \overline{\text{span}\{u_i \otimes v_j : i, j \in \mathbb{N}\}},$$

which means that, in this scenario, the tensor product can be naturally identified with the closure of the space of finite linear combinations of elements $u \otimes v$, where $u \in \mathcal{H}_1$ and $v \in \mathcal{H}_2$, which are called *simple tensors* or *pure tensors*. Moreover, the set of simple tensors with factors taken from the respective complete orthonormal systems forms a complete orthonormal system for the tensor product.

Remark 31. With a simple inductive process one can conclude that the N-fold tensor product of Hilbert spaces is a Hilbert space, and the closure of the set of finite linear combinations of simple tensors still coincides with that tensor product.

Remark 32. A useful note for the type of computations we want to tackle in this work is that the space $L^2(\mathbb{R}^n) \otimes L^2(\mathbb{R}^m)$ can be naturally identified with the space $L^2(\mathbb{R}^{n+m})$, and given $f \in L^2(\mathbb{R}^n)$ and $g \in L^2(\mathbb{R}^m)$ one has that

$$f \otimes g(x, y) = f(x)g(y).$$

Lastly, we may define operators on a tensor product starting from operators on the factors. Indeed, let $A : D(A) \subseteq \mathcal{H}_1 \rightarrow \mathcal{H}_1$ and $B : D(B) \subseteq \mathcal{H}_2 \rightarrow \mathcal{H}_2$ be operators on the respective Hilbert space, then we can define their tensor product as

$$(A \otimes B)(\phi \otimes \psi) := (A\phi) \otimes (B\psi)$$

on the set of pure pure tensors, i.e.

$$D(A \otimes B) = \{\phi \otimes \psi : \phi \in D(A), \psi \in D(B)\}.$$

[1]

References

- [1] What. “wHAT”. In: *wHaT* (which year again?).